Government of Nepal
Ministry of Energy, Water Resources and Irrigation
Department of Water Resources and Irrigation
Irrigation Management Division
System and Integrated Crop Water Management Program

A Final Report
on
Preparation of Canal Operation Plan of Kamala Irrigation System (Eastern and Western Canal)


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Acronyms

AKC	Agriculture Knowledge Center
CCA	Cultivated command area
COP	Canal Operation plan
CWR	Crop water requirements
DWRI	Department of Water Resources and Irrigation
EMC	Eastern main canal
FAO	Food and agriculture organization
FFD	First Fit Decreasing
GCA	Gross command Area
GIS	Geographic Information System
Ha	Hectares
IDR	Irrigation Diversion Requirements
IMD	Irrigation Management Division
IWR	Irrigation water requirements
IWRMP	Integrated watser resource management plan
KIS	Kamala Irrigation System
NCA	Net command area
TOR	Terms of Reference
USDA	United States Department of Agriculture
WMC	Western main canal
WUA	Water Users Association

1. Introduction

1.1 Project background

Department of Water Resources and Irrigation (DWRI) is a government organization, with a mandate to plan, develop, maintain, operate, manage and monitor different modes of environmentally sustainable and socially acceptable Irrigation Projects both surface and ground water or projects of water resources with irrigation as major component, particularly larger in size. Many of these systems are being managed jointly by the DWRI and respective WUAs and the performance of these systems, however, have remained below the regional average. Some attempts have been made to enhance the performance of these irrigation systems through Participatory Irrigation Management or Irrigation Management transfer. Despite continuous efforts to improve the system performance there are several reasons associated with the dismal performance of Irrigation System.

Realizing these circumstances, the DWRI under the initiation of Irrigation Management Division (IMD) has stepped up with series of integrated action plans to support the improvement of irrigation system management ensuring sustainable operation and hence enhance the system efficiency. Improvement of canal operation and their sustainable operation require number of plans and their effective implementation. Among several steps of improving canal and system performance, Assets Inventory, Asset Management Plan, Parcellary Map, Canal Operation Plan and Calibration of Structures are some of utmost areas that need consideration.

Besides this, for enhancing canal and system performance, it is important to develop a Canal Operation Plan (COP) within irrigation systems. The COP is described as an operational schedule for a canal, specifying the timely and spatial delivery of irrigation water in terms of location, flow rate, delivery time, and duration. It covers activities such as water acquisition, allocation, and distribution, aiming to establish a systematic approach for delivering reliable and equitable irrigation water to farmers.

Recognizing the importance of COP and in response to a request from the KIS office, the IMD has initiated the work for the development of Canal Operation Plan for KIS with an aim of improving the irrigation service performance and delivery to KIS users. The COP must be developed for the main canal of KIS, branch and distributaries for all cropping (Irrigation) Season. The TOR encompasses all the works related to the development of COP for KIS, and if any task that falling under task's scope or objective has been previously completed the results shall be reviewed and integrated to fulfil task's objective. The TOR includes the information regarding the existing documents to extent possible but is the responsibility of the consultant to investigate, review and conduct necessary inquiry about previous studies.

1.2 Objectives

The overall objective of the assignment is to formulate COP for Kamala Irrigation System keeping in view of the cropping pattern, water demand in the command area. The specific objectives of COP Works, but are not limited to the following:

- To evaluate the existing irrigation system including existing irrigation practice for all seasons.
- To Assess the irrigable area at the level of each farm gate (at tertiary level) and compile them at the level of canal head end.
- To carry out agricultural surveys within distributaries study area and estimate the demands of each cropping season.
- To verify discharge capacity of canal at appropriate locations and access the dependable water availability at the intake and at farm gates using the approved efficiency factors
- To evaluate the water balance of the irrigation system.
- To develop water delivery scheduling to different possible crops for study area, its branch and distributaries for all season.
- To develop COP for study area, its branches and distributaries for all cropping season.
- To suggest remedial measures for the canal system to enhance the performance of water delivery services.

1.3 Scope of the work

In order to achieve the above objectives, the scope of the work for this assignment is, but not limited to:

Table 1 : Scope of the assigned work

Scope	Activities
Collection and review of existing documents and literatures	To collect all the available project related reports, study reports including maps and drawings, guidelines and other information of KIS. Previous study on Asset Inventory and Irrigation Area Boundary Mapping of KIS can was acquired from Irrigation Management Division, Jawalakhel.
Inventory of Hydraulic Structures	To conduct walkthrough to identify outlets, authorized and un-authorized, in order to consider delivery of bulk water to respective branch, distributary and secondary canals. The survey would mainly cover, but not limited to, the following activities: • Identification of outlets and dimensions along KIS, branches and distributaries. • Assess the functionality of existing hydraulic structures

Scope	Activities
	• Assessment of the cross regulator to maintain depth of water in canal to meet the duty of different crops for proper water management and also to interpret and analyze the field data and prepare the list and report as required for COP works.
Agricultural Survey Works	To list existing and possible crop data, crop and crop rotation pattern for the command area in consultation with respective WUA and lead farmers. Consult with respective agriculture office from Agriculture Knowledge Center (AKC) or other relevant agency in order to efficiently assess cropping pattern and crop yields within the KIS command area. Nature of existing soils predominating over command area should be coupled with proposed crop types and water management practice. To prepare cropping pattern separately for each branch canals of KIS in order to assure delivery of bulk water to respective branch canals and its distributaries and secondary canals.
Assessment of canal capacity and discharge record at KIS head	To measure the discharge carrying capacity at the head of branch and distributary canals in order to match with proposed bulk water delivery using current meter or other appropriate tools and techniques. Further, the consultant would assess the past measured discharge values at the head of KIS using appropriate statistical methods for balancing seasonal crop water requirements with dependable flow through head of KIS.
Preparation and Development of COP	To estimate Crop Water Requirement (CWR) using CROPWAT model. The output of CWR must be balanced with dependable flow at the head of KIS that leads to water delivery schedule of respective secondary canals, branch canals and KIS. To prepare COP that should also propose canal operation procedure together with clear roles and responsibility of WUA, gate operators, and engineering professionals' in-charge of canal operation and management of respective canal system.

2 Approach and Methodology

2.1 Approach

The Canal Operation Plan (COP) for the Kamala Irrigation Project has been developed using a practical, participatory, and phased approach that prioritizes reliable bulk water delivery from the main canal while ensuring efficient and equitable distribution at the branch, minor, and tertiary levels. This strategy follows the guidelines within the IWRMP, tackling typical issues in Terai region of Nepal like unfair water sharing, lack of water at the end of canals, significant losses, and poor organizational links.

The main strategy uses supply-focused water delivery scheduling as its core operational approach. This option prioritizes ease of use, simplicity, and dependability, which is crucial for Water Users' Associations (WUAs) overseeing smaller canals. Scheduling based on supply relies on the water availability at the source, the discharge capacities of canals, and predetermined delivery schedules. It reduces operational complexity and the need for frequent gate adjustments compared to fully demand-oriented systems, which are difficult with current infrastructure and management capacity.

Considering that supply-driven operations have limitations, such as significant water wastage and a disconnect from actual crop requirements, the method allows for progressive modernization. With stronger WUA capacity resulting from better communication, improved demand forecasting by tertiary committees, and consistent feedback, the system can move towards scheduling that better meets demand. This combined approach allows for quick implementation while offering future improvements in water usage efficiency, possibly leading to substantial water savings over conventional methods.

Table 2 : Key guiding principles of the approach

Principle	Key Focus
Reliability & Predictability	Advance water schedule shared so farmers can plan activities
Equity	Fair distribution, priority to tail-end farmers, controlled withdrawals
Efficiency	Optimal water use, reduced losses, proper canal operation
Participatory Management	Involvement of WUAs and stakeholders in decision-making
Sustainability	Link O&M, cost awareness, and capacity building
Safety & Protection	Controlled flow changes, safe gate operation, avoid crop damage from sudden and over water supply

The approach is structured around the following main components:

1. System Assessment: Thorough understanding of the existing physical infrastructure, current operational status, soil data, and existing cropping patterns.
2. Irrigation Planning: Calculation of water requirements, water balance analysis, and preparation of seasonal water delivery schedules.
3. Institutional and Operational Arrangements: Clear definition of roles between the agency and WUA, standardized procedures for gate operation, communication, patrolling, and enforcement of regulations.

2.2 Methodology

This Canal Operation Plan has been prepared following a systematic, participatory, and technically sound approach to ensure reliable, efficient, and equitable water delivery in the Kamala irrigation system. The methodological framework follows an integrated seven phase methodology that combines data collection, data analysis using various software, system assessment, water balance evaluation, and scheduling. This approach ensures technical soundness, operational feasibility, and stakeholder acceptability while maintaining iterative refinement throughout the process.

Table 3 : Phase wise activities for COP preparation

Phase	Key Activities
1. Data Collection	Existing Documents & Literature, Physical System Data, Soil Data, Climate Data, Agricultural Data, Hydraulic Data
2. Data Analysis	Spatial & Infrastructure Analysis (GIS), Irrigation water requirements and Irrigation scheduling Analysis (CROPWAT), Hydraulic & System Analysis, Stakeholder Consultation (Continuous)
3. System Assessment	Overall System Assessment (Integration Layer): Canal network inventory, Command area assessment, WUA characterization, Structure performance evaluation, Data gap analysis
4. Water Assessment	Dependable flow analysis, Canal carrying capacity, Conveyance efficiency, Available water at control points
5. Water Balance	Core Decision Node: Demand vs Supply, Seasonal surplus/deficit, Head-tail equity assessment, Loss estimation, Water prioritization
6. Scheduling	Water delivery schedules, Continuous supply, Rotational supply, Duty & delta determination, Gate operation rules
7. Final Output	Canal Operation Plan (COP): Branch-wise water allocation, Irrigation schedules, Cropping recommendations, Canal operation procedures, Institutional roles & responsibilities

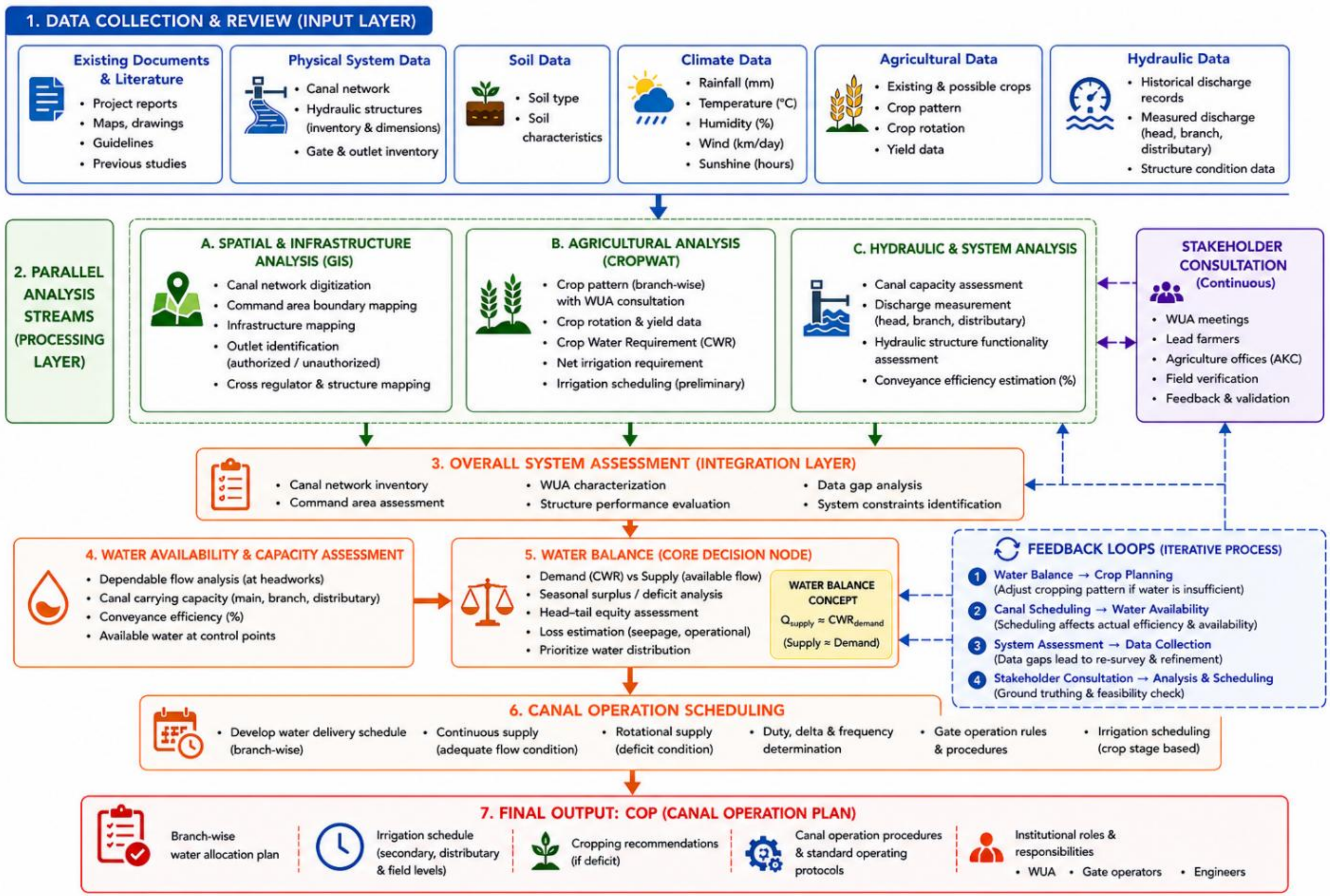


Figure 1 : Overall methodological framework for preparation of COP

2.2.1 Phase 1: Data Collection and Review (Input Layer)

The first phase in the preparation of COP involves data collection pertaining hydraulic, hydrologic, physical, agricultural and socioeconomical domains and review from multiple sources. This input layer establishes the foundation for all subsequent analyses by ensuring availability of accurate, current, and complete information regarding the irrigation system and its context.

1. Review of Existing Documents and Reports

A comprehensive review of existing documentation should be done to build a clear understanding of the Kamala irrigation system. Project reports, including feasibility studies and design documents, were examined to assess the system background and current condition. Asset inventory records were used to gather detailed information on canal structures and their specifications. In addition, relevant irrigation design standards and operational guidelines were reviewed to ensure alignment with best practices. Previous studies on cropping patterns and agricultural productivity were also analyzed to better understand existing agricultural activities in the command area.

Table 4 : Categories of reviewed documents and reports

Category	Description
Project Reports	Reviewed feasibility studies, design reports, and planning documents to understand system background and conditions.
Asset Inventory	Consulted past inventory reports to gather details on canal structures and specifications.
Design Guidelines & Standards	Examined irrigation standards and operational guidelines for best practice compliance.
Cropping Pattern Studies	Analyzed previous studies to understand agricultural practices and productivity.

2. Inventory of Physical System

A detailed inventory of the canal system was prepared through compilation of existing records. These data and other unrecorded data were validated and collected through field surveys. The inventory of Physical system included the following:

Table 5 : Data collected for physical system inventory

Assessment Component	Key Information Collected
Canal Network Structure	Identification and mapping of main, branch, minor canals, including spatial distribution and hierarchical classification.
Canal Characteristics	Documentation of canal length, type (lined/unlined) design discharge capacity (m^3/s), and command area (ha) for each canal segment.

Assessment Component	Key Information Collected
Hydraulic Structures	Inventory of head regulators, cross regulators, escape channels, drop structures, bridges, other cross-drainage works, including dimensions, condition, and functionality.
Gate Conditions and Status	Recording of operational status, mechanical condition, ease of operation, and defects of all gates.
Unauthorized Interventions	Observation and documentation of illegal cuts, pipe insertions, barriers, and other unauthorized modifications affecting canal operation and water distribution.

3. Hydraulic data inventory with field Observations

Multiple field visits were conducted at that period when canal is under operation to observe actual flows, operational practices and conditions. Key observations included the followings:

Table 6 : Areas of Hydraulic data inventory

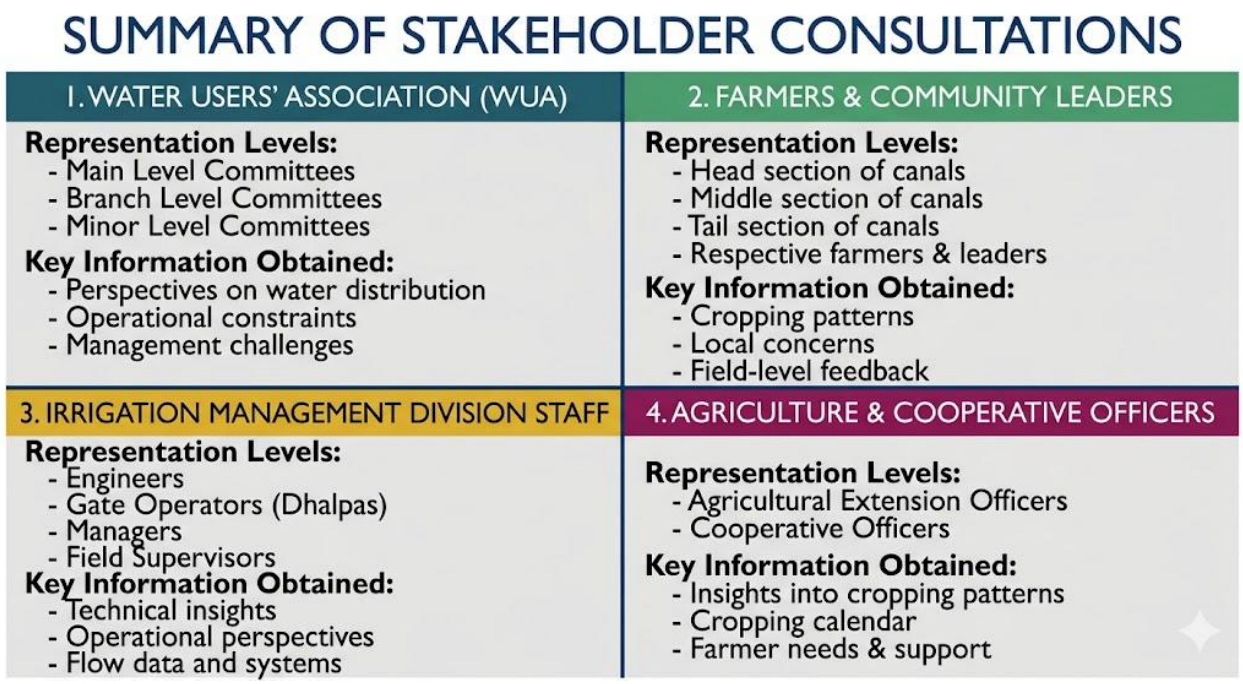
Observation Area	Key Information Observed
Available discharge	Water flow at the gates of each tier canal network i.e. Main canal, Branch canal and Minor canals
Canal Operation Practices	Observation of gate operation procedures, water level management, and seasonal control practices.
Reach-wise Problems	Identification of issues at head, middle, and tail reaches, including inequitable distribution, farmer dissatisfaction, and infrastructure deterioration.
Farmer Behaviors and Informal Practices	Recording of unauthorized pipe insertions, illegal canal cuts, barriers, and other farmer interventions affecting water flow.

4. Stakeholder Consultations

Structured consultations were conducted with various stakeholder groups at different levels:

- Water Users' Association (WUA) at All Levels: Consultations were held with the WUA at main and branch level committee levels to understand their perspectives on water distribution, operational constraints, and management challenges.
- Farmers: Individual meetings with respective farmers of all canal levels at head, middle and tail section of that canal and community leaders helped identify cropping pattern, local concerns and gather field-level feedback.

- Irrigation Management Division Staff: Discussions with engineers, gate operators (Dhalpas), and managers of the Irrigation Management Division provided technical insights and operational perspectives.
- Agriculture and Cooperative Officers: Consultations with agricultural extension and cooperative officers provided insights into cropping patterns, cropping calendar, farmer needs etc.



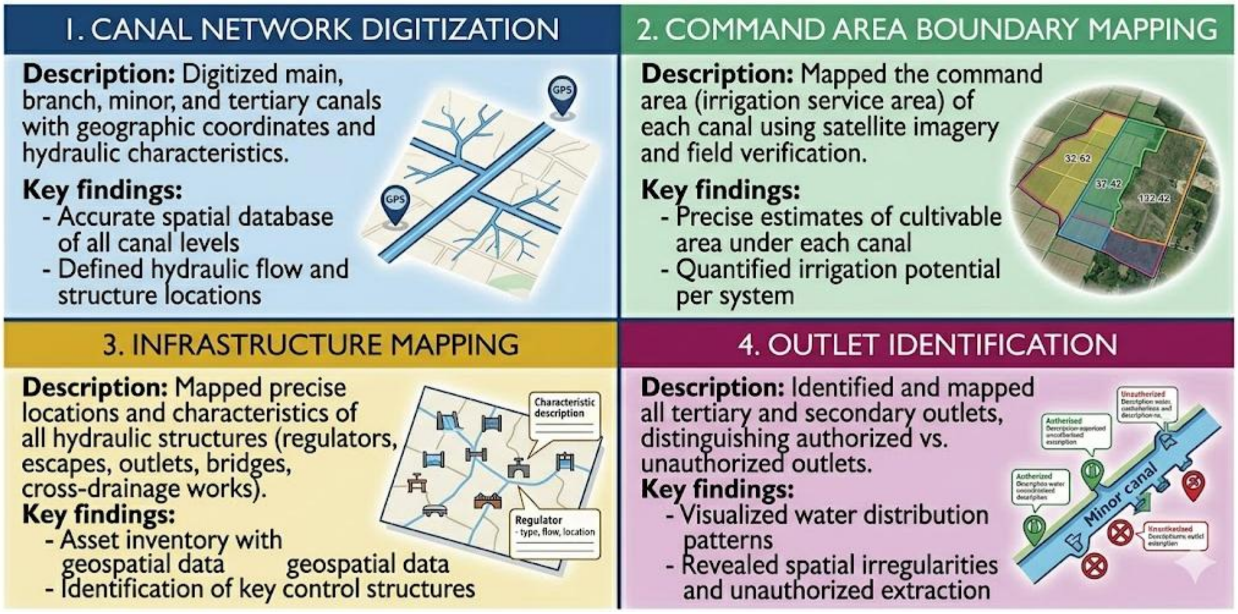
2.2.2 Phase II: Data Analysis

1. Spatial and Infrastructure Analysis (GIS Stream)

Geographic Information System (GIS) analysis was used using Arc Map 10.2, to map and visualize the spatial dimensions of the irrigation system.

Table 7 : Activities performed using GIS

GIS Mapping Activity	Key Output
Canal Network Digitization	Spatial database of main, branch, minor canals, and tertiary outlets with geographic and hydraulic details.
Command Area Boundary Mapping	Mapping and land use classification of canal command areas using satellite imagery and field verification.
Infrastructure Mapping	Location mapping of regulators, escapes, outlets, bridges, and cross-drainage structures with key characteristics.
Outlet Identification	Identification and mapping of authorized and unauthorized tertiary and secondary outlets to assess water distribution patterns.



2. Agricultural Analysis (CROPWAT Stream)

The FAO CROPWAT 8.0 model was utilized for comprehensive agricultural and water requirement analysis. The analysis resulted in following outputs:

Table 8 : Analysis component of CROPWAT results

Analysis Component	Key Output
Crop Pattern Analysis	Existing and proposed crops, cropping intensity, and rotation practices for each branch command area.
Crop Water Requirement (CWR)	Estimation of crop water needs using the CROPWAT model with climatic, soil, crop, and rainfall data.
Net Irrigation Requirement (NIR)	Calculation of irrigation demand after deducting effective rainfall from crop water requirement.
Gross Irrigation Requirement (GIR)	Estimation of total water delivery requirement considering field application efficiency.
Land Preparation Requirement	Assessment of paddy and other major crops soaking, puddling, standing water, and percolation losses during land preparation.
Irrigation Scheduling	Development of irrigation timing and water application schedules for optimal crop production.

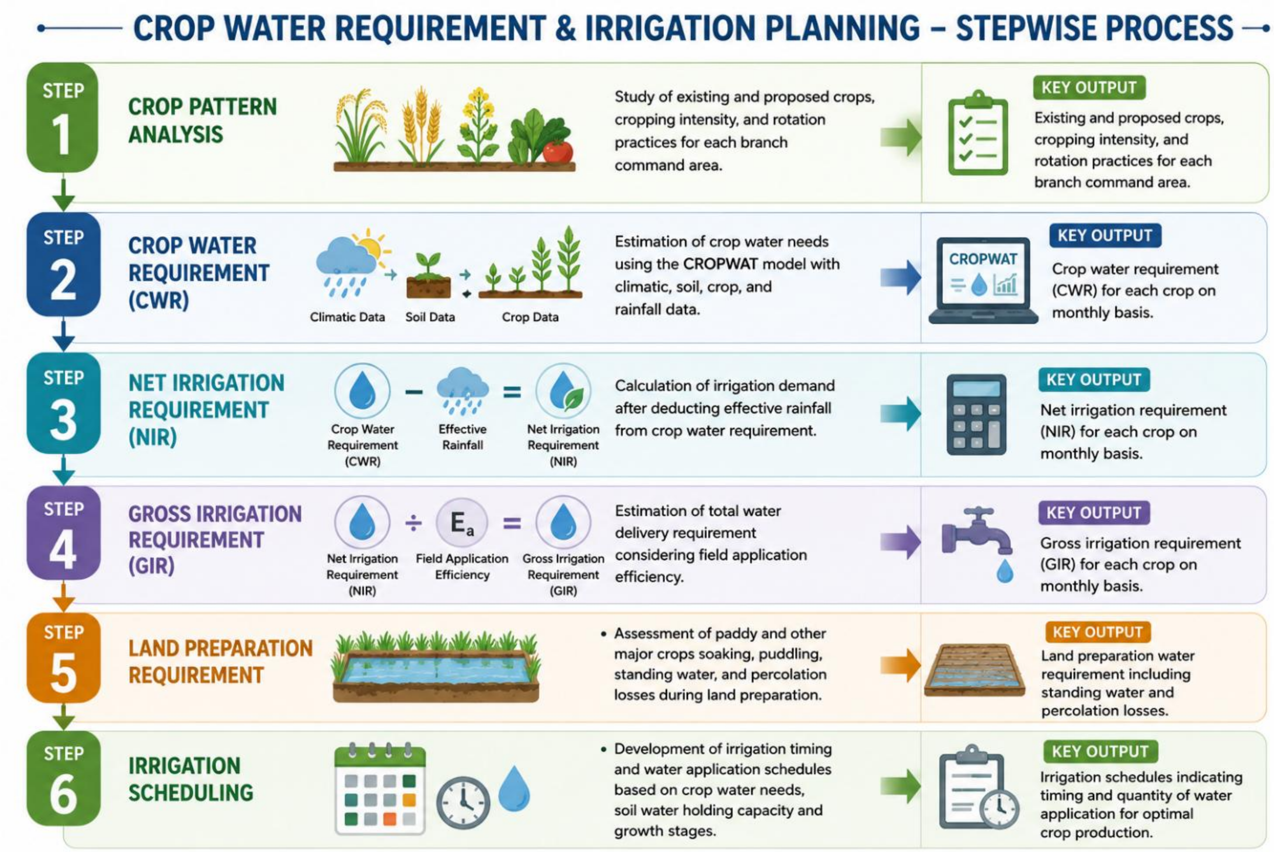


Figure 2 : Irrigation planning method adopted

3. Hydraulic and System Analysis

Detailed hydraulic analysis was conducted to assess the capacity and functionality of the canal system.

Table 9 : Assessment component of Hydraulic and System analysis

Assessment Component	Key Output
Canal Capacity Assessment	Evaluation of carrying capacity considering design capacity, siltation, encroachment, and deterioration.
Discharge Measurements	Measurement of actual canal discharge at key control points to compare with design capacities.
Hydraulic Structure Assessment	Assessment of regulators, escapes, and outlets for operational performance and structural issues.
Conveyance Efficiency Estimation	Estimation of water losses due to seepage, percolation, and evaporation, and calculation of conveyance efficiency.

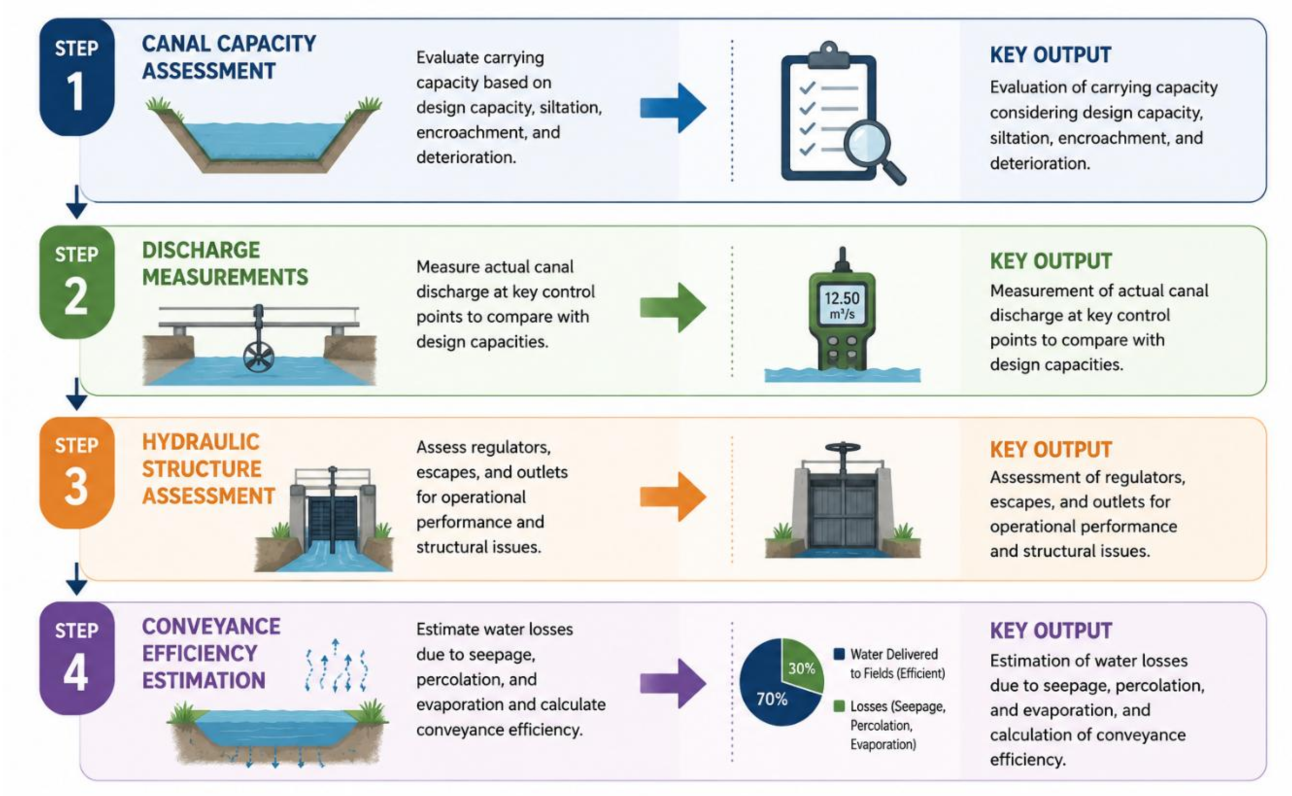


Figure 3 : Process of Hydraulic and system analysis

2.2.3 Phase III: Overall system Assessment

The overall system assessment integrated findings from spatial, agricultural, and hydraulic analyses to develop a comprehensive understanding of the irrigation system. A complete canal network inventory was prepared by combining information on canal location, command area, hydraulic capacity, structural condition, and operational features such as gates and outlets. Command areas for each canal section were assessed in terms of land use, cropping patterns, farmer groups, and organizational arrangements. The functionality and institutional capacity of Water Users' Associations (WUAs) were evaluated, including their technical knowledge, administrative performance, coordination, and acceptance among farmers. Major hydraulic structures were assessed based on their operational performance and categorized as functioning well, partially functioning, or non-functional. In addition, data gaps were identified, including areas requiring further field measurements and additional surveys. Based on the integrated assessment, key system constraints affecting equitable water distribution and efficient operation were identified, including inadequate hydraulic head for tail-endss irrigation, excessive seepage losses, poorly maintained hydraulic structures, weak WUA coordination, unauthorized water extraction, and limited water availability during the dry season.

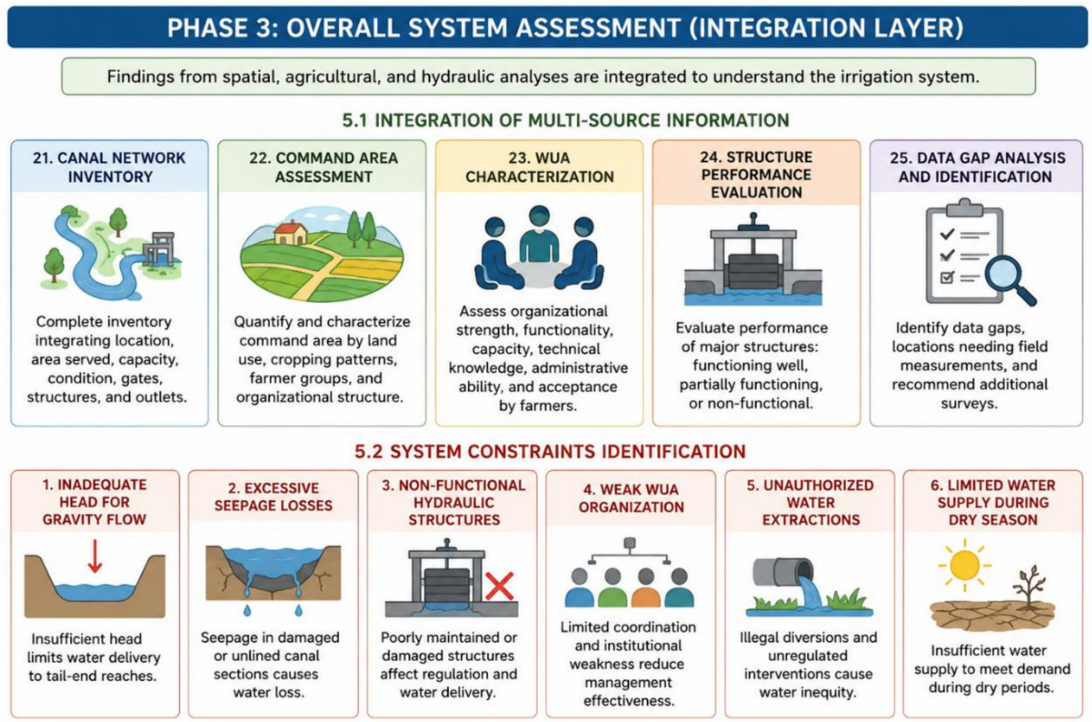


Figure 4 : Process of overall system assessment

2.2.4 Phase IV & V: Water availability and balance analysis

a) Water Availability Assessment

This phase assesses the quantity of water available from the source and its seasonal variation. Following set of steps were followed:

1. Dependable Flow Analysis: Long-term hydrological data for the kamala river source was analyzed to determine the dependable flow (flow available with specified reliability, typically 75% probability) at the headworks.
2. Seasonal Flow Variation: Flow data was analyzed to determine monthly and seasonal variations, identifying surplus months (monsoon) and deficit months (dry season).
3. Water Diversion Records: Actual water diverted into the system from the source was documented for different seasons based on available operation records and measured discharge.
4. Available Water at Control Points: Available water at each control point (head regulator, branch, minor gates) was determined by accounting for upstream conveyance losses, accounting for the conveyance efficiency calculated in Phase 2.

b) Water Balance Development

Water balance was prepared by following the set of given steps which was the basis for decision making on water allocation and scheduling. The water balance compared water supply with water demand on a monthly basis.

1. Demand Determination: Total monthly irrigation demand was determined by aggregating the gross irrigation requirements of all crops in each command area, derived from the CROPWAT analysis in Phase 2.
2. Supply Assessment: Available water supply from the source, adjusted for conveyance losses, was determined on a monthly basis.
3. Water Balance Computation: Water balance (Supply minus Demand) was computed for each month, identifying periods of surplus and deficit.
4. Surplus and Deficit Analysis: Months with surplus water indicated when continuous could be maintained, while deficit months revealed the need for rationing or rotational distribution.
5. Head-Tail Equity Assessment: Special attention was given to analyzing how water shortage would affect the head reach (near intake or Head regulators) and tail reach (far from intake or Head regulators) farmers. Measures to ensure equitable distribution during shortage were identified.
6. Loss Estimation and Prioritization: Conveyance and field application losses were quantified to understand the total water requirement from the source. Strategies to reduce losses through canal lining, maintenance, and improved irrigation efficiency were identified and prioritized.

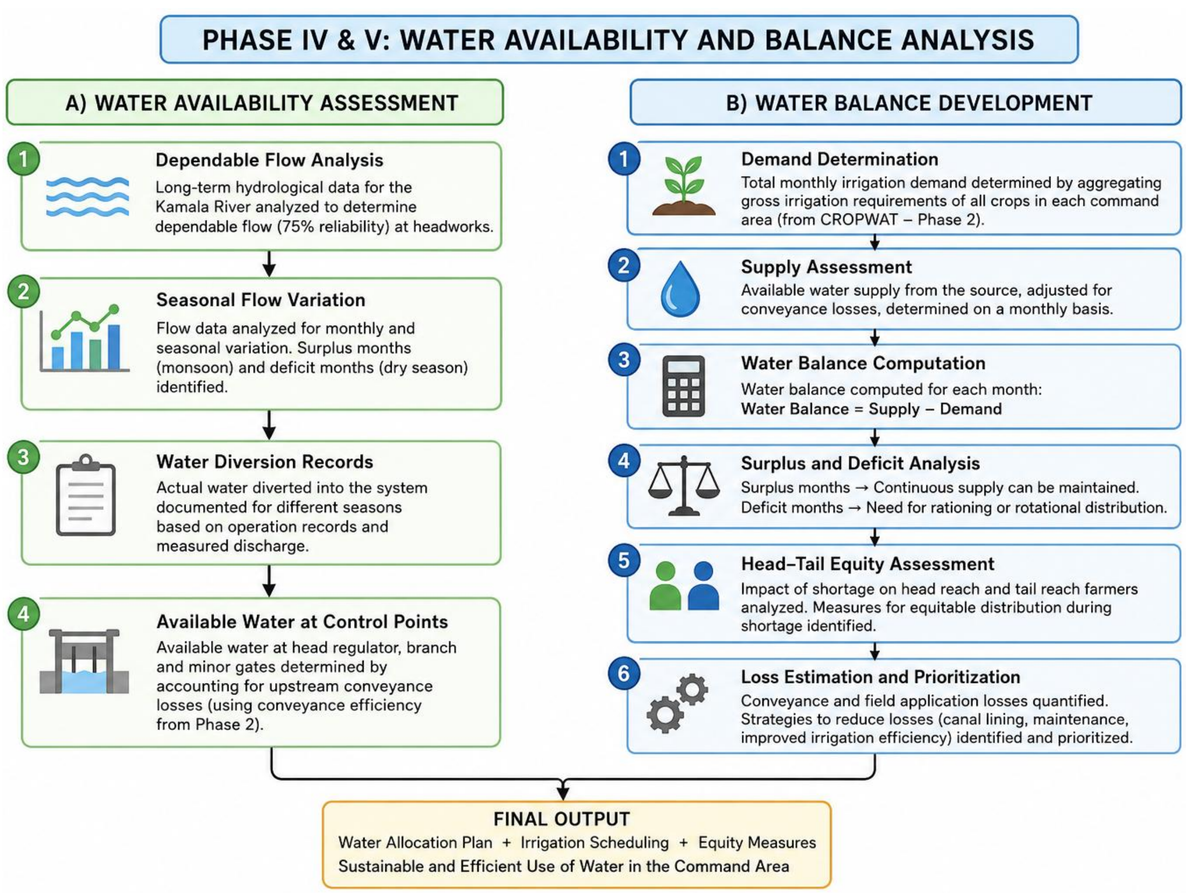


Figure 5 :Water availability and Balance analysis procedure

2.2.5 Phase VI: Canal Operation scheduling

Based on the water balance analysis, detailed water delivery schedules were developed to translate water availability into operational procedures.

1. Development of Water Delivery Schedules

Water delivery schedules were prepared using a supply-oriented approach, designed to be simple and operational for implementation while ensuring equitable distribution.

Wet Season (Monsoon) Schedule: During monsoon, when water availability is typically abundant, a continuous supply schedule was designed. All branch and minor canals receive water continuously at or near design discharge, based on the major crop's irrigation water requirement. The duration of the delivery was calculated by incorporating the irrigation water requirement (in mm), flow of water available (l/s) and area under irrigation (ha) of irrigation canal under consideration.

Dry Season (Winter & Spring) Schedule: During dry seasons, if there is sufficiency of irrigation water, a continuous approach as that of monsoon season was applied. However, when water is limited, a rotational supply schedule was designed. Branch and minor canals were grouped into rotation groups of either approximately equal cumulative design discharge or Command area. Each group receives water supply for a fixed rotation period with duration of the delivery calculated by incorporating the irrigation water requirement (in mm) , flow of water available (l/s) and area under irrigation (ha) of irrigation canal under consideration

2. Development of Irrigation schedules

Grouping for Rotation: Branch/minor canals were grouped either by dividing the total design discharge into rotation groups of approximately equal cumulative discharge, or based on proportionate distribution of area to be irrigated (in Ha).

I. Rotation Roster Development: For each rotation group, a rotation roster was prepared showing:
    - (a) the specific canals included in the group,
    - (b) the cumulative design discharge of the group,
    - (c) the assigned rotation days per week,
    - (d) the hours of supply per day, and
    - (e) the sequence and timing of rotations.

II. Duty and Delta Determination: The concept of 'duty' (area that can be irrigated from a unit discharge) and 'delta' (depth of water required for seasonal crop production) were calculated for major crops using CROPWAT software (Phase II). These parameters guided water allocation decisions and verification of adequacy of water supply for planned cropping patterns.

III. Frequency of Supply: The frequency of water supply to each rotation group (e.g., once per 7 days, once per 5 days) was determined based on crop water requirements and soil water-holding capacity, aiming to match irrigation intervals to crop needs.

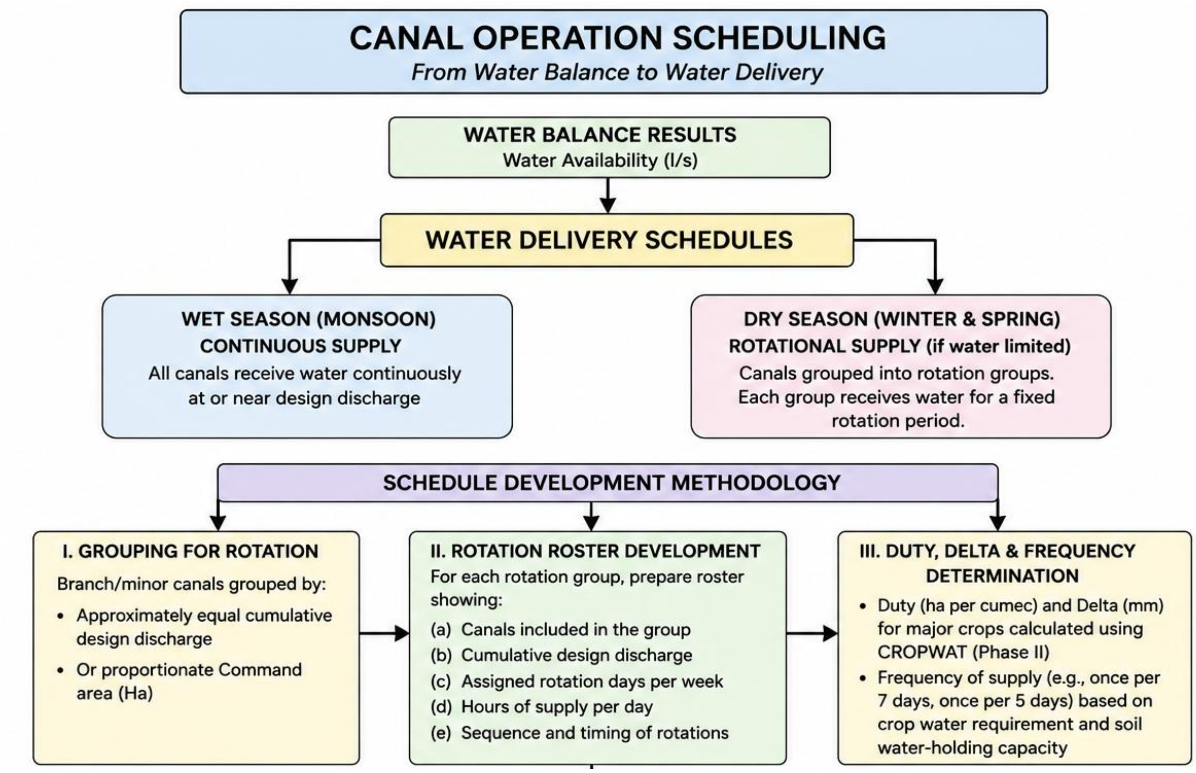


Figure 6 : Canal operation scheduling process

3 Description of the Physical System

3.1 Project Background and Location

The Kamala Irrigation System (KIS) is a key to agricultural development in the Madhesh Province of Nepal. Spanning the banks of the Kamala River and extending across the boundary between Dhanusha and Siraha Districts, was established in 1960 AD. Accessible via the east-west highway, which intersects the head of its command area, the KIS serves as a lifeline for local farmers. The backbone of the KIS comprises two main canals: The Eastern Canal and the Western Canal. These canals run parallel to the Kamala River, serving as conduits for diverting water to various agricultural fields within the command area. The Eastern Canal, with its extensive network of branches and minors, effectively distributes water across vast swathes of farmland, ensuring optimal irrigation coverage. Similarly, the Western Canal complements this network, further enhancing water distribution efficiency and agricultural productivity in the region.

Over the years, the KIS has undergone numerous upgrades and expansions to meet the evolving needs of local farmers and keep pace with technological advancements in irrigation management. From the installation of modern control systems to the implementation of sustainable water management practices, the system has continually evolved to maximize its efficiency and effectiveness. Moreover, the system empowers them to increase crop yields, improve livelihoods, and contribute to overall community prosperity.

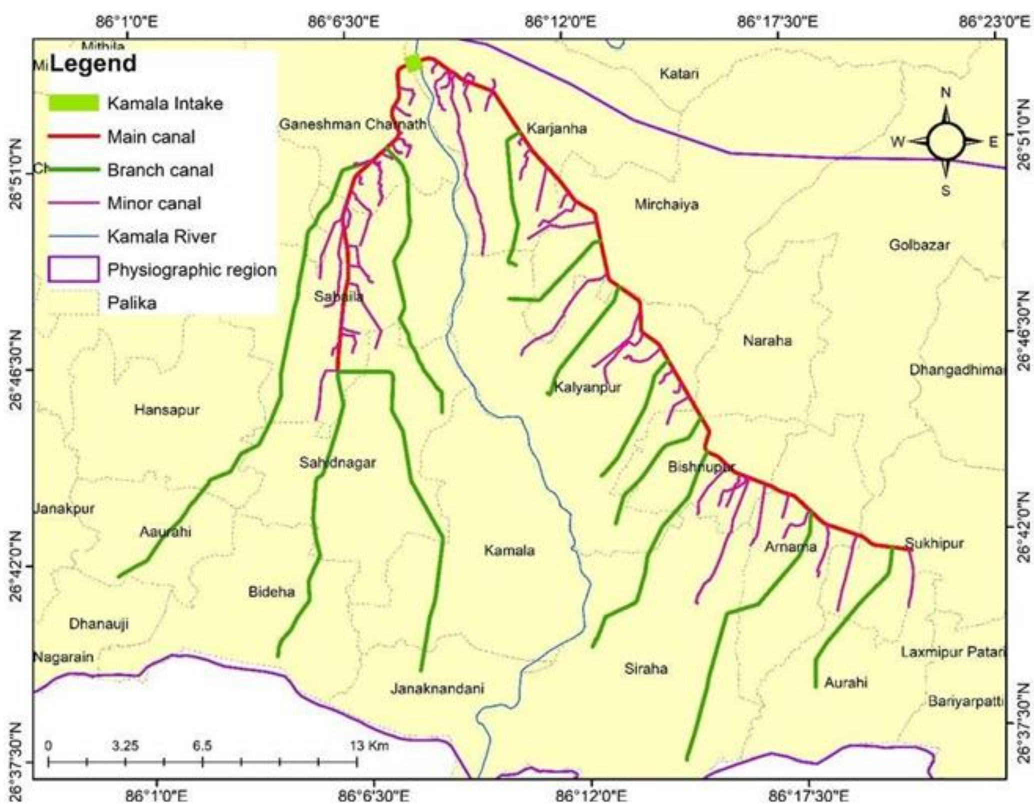


Figure 7 : Canal network distribution of Kamala Irrigation System

3.2 Inventory of Canal Structures

The canal network of KIS comprises of two main canal i.e East main canal toward Siraha District having length of 31.8 km and West main canal toward Dhanusha District of length 14.2 km. Design capacity of both East and West main canal is 14 m^3/s which is the available water of the river during monsoon.

3.2.1 Eastern Canal System

In East main canal, there are eight branch canals of length between 4.9 km of Sikron branch and maximum of 12.5 km of Lagadhi branch. There are 32 minor which is distributed from east main canal. The main canal is lined and unlined at different places.

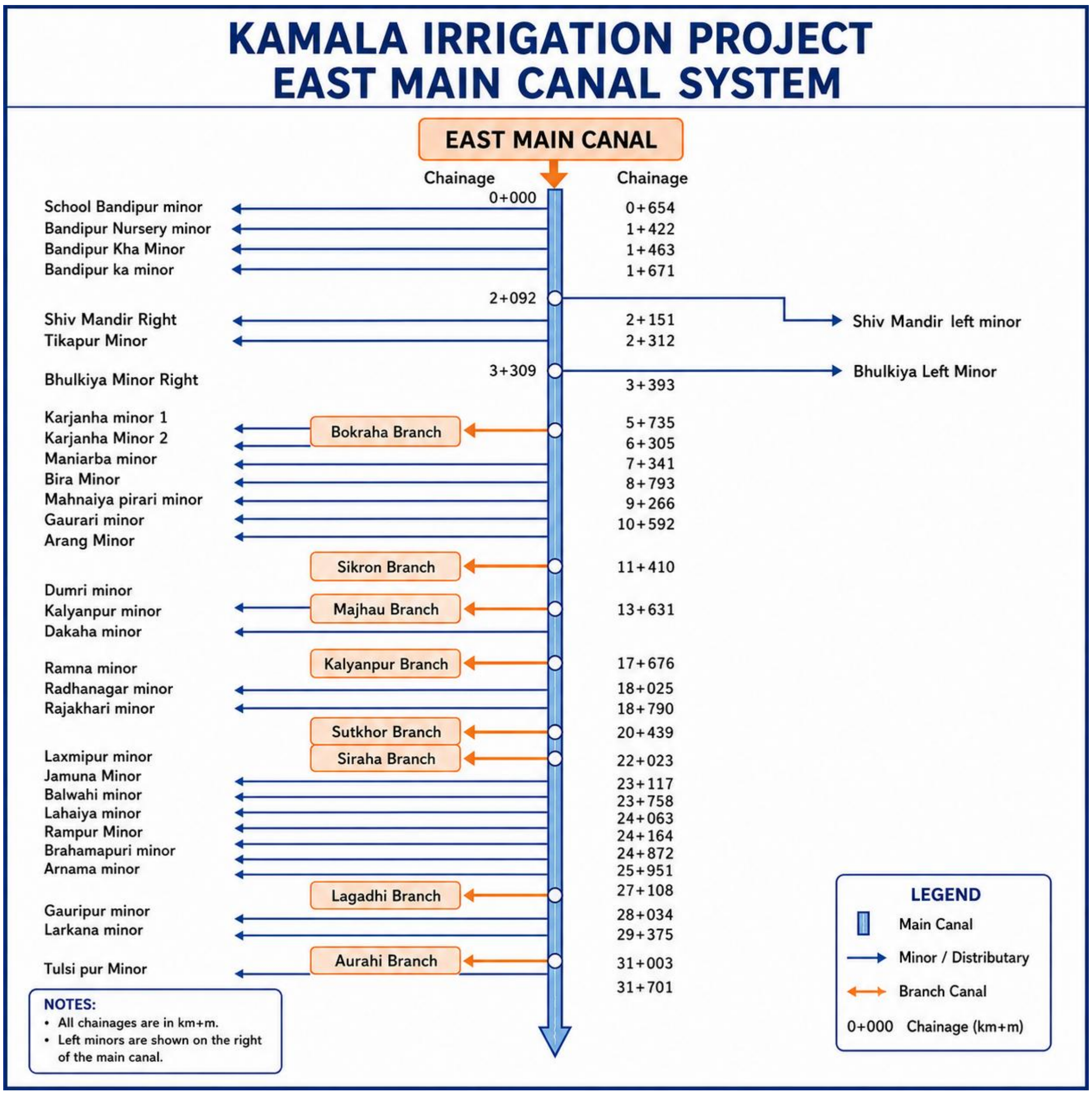


Figure 8 : Canal layout diagram of KIS (EMC)

Table 10 : EMC Canal Properties

S.no.	Name	Type	length (Km)
1	Bokraha Branch	Branch	6.2
2	Sikron Branch	Branch	4.9
3	Majhau Branch	Branch	5.6
4	Kalyanpur Branch	Branch	5.7
5	Sukhathor Branch	Branch	6.1
6	Siraha branch	Branch	10
7	Lagdhi Branch	Branch	12.5
8	Aaurahi Branch	Branch	8
9	School Bandipur minor	Minor	0.6
10	Bandipur Nursery minor	Minor	2.4
11	Bandipur Kha Minor	Minor	1
12	Bandipur ka minor	Minor	8
13	shiv mandir left minor	Minor	0.9
14	Shiv Mandhir Right	Minor	0.2
15	Tikapur Minor	Minor	1.3
16	Bhulkiya Left Minor	Minor	0.6
17	Bhulkiya Minor Right	Minor	0.9
18	Karjanha minor 1	Minor	0.4
19	Karjana Minor 2	Minor	1.6
20	Manharba minor	Minor	1.9
21	Birta Minor	Minor	0.7
22	Malhaniya pirari minor	Minor	2.1
23	Gautari minor	Minor	3.1
24	Arang Minor	Minor	5.3
25	Dumri minor	Minor	3.3
26	Kalyanpur minor	Minor	3.2
27	Dakaha minor	Minor	2
28	Ramna minor	Minor	0.9
29	Radhanagar minor	Minor	0.6
30	Rajokhari minor	Minor	1.6
31	Laxmipur minor	Minor	2.5
32	Jamuwa Minor	Minor	1.6
33	Balwahi minor	Minor	1.9

S.no.	Name	Type	length (Km)
34	Larhaiya minor	Minor	6
35	Rampur Minor	Minor	2.7
36	Brahamapuri minor	Minor	1
37	Arnama minor	Minor	1.9
38	Gauripur minor	Minor	2.4
39	Larkana minor	Minor	3.3
40	Tulsipur Minor	Minor	2.5

3.2.2 Western Canal System

In West main canal, there are four branch canals of 12.3 km Raghunathpur branch to largest Parwaha branch canal 22 km and a total of 24 minors. The canal network diagram of EMC shown below.

The bed width of the main canal is 6 m. and right bank of this canal is 2m wide whereas left bank has 4m width. The Left bank serves as service road for West main canal for inspection and travel purposes for the office and villagers. For East main canal, left and right bank both serve as service road.

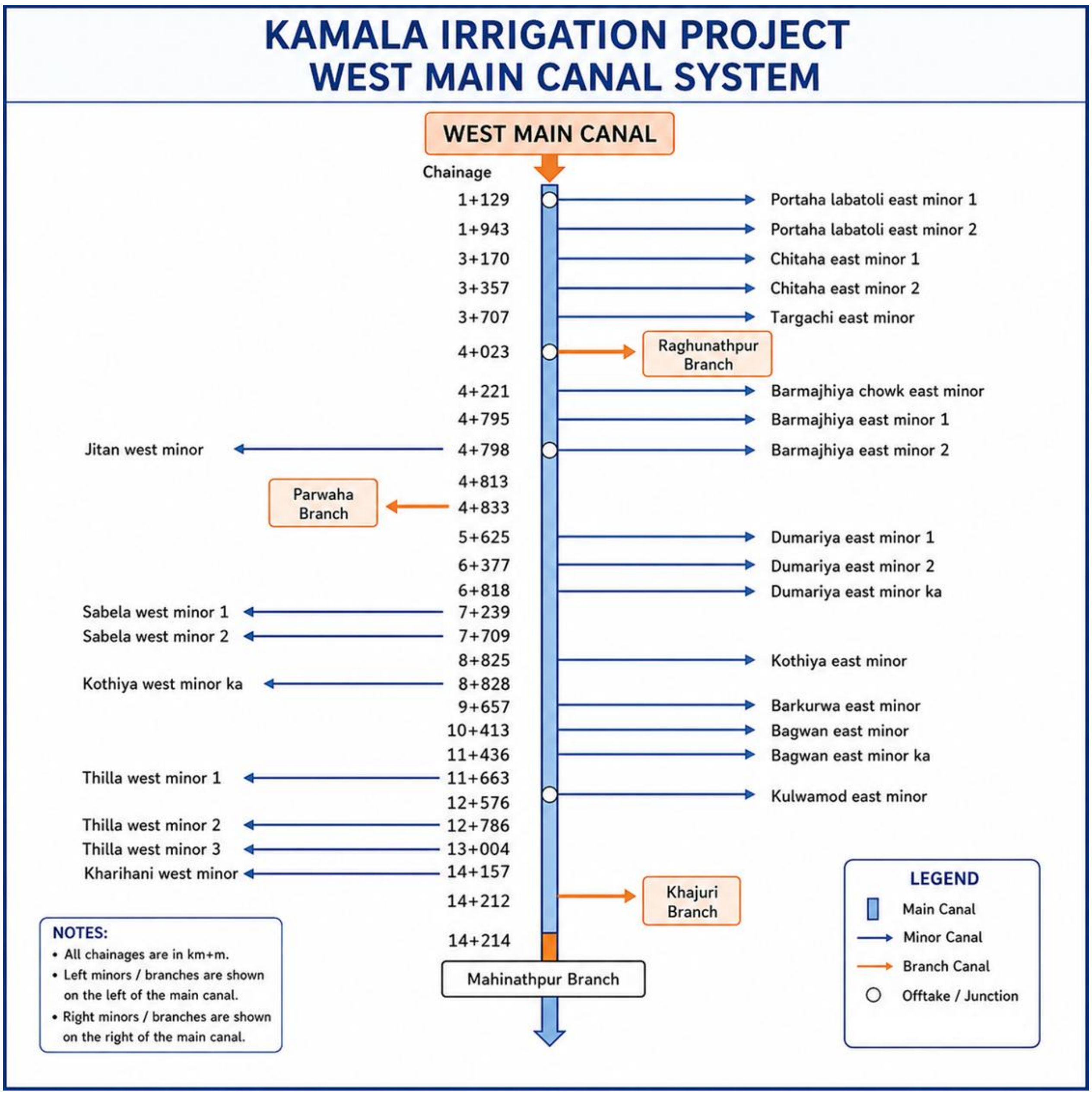


Figure 9 : Canal layout diagram of KIS (WMC)

Table 11: Canal layout diagram of KIS (WMC)

S.no.	Name	Tier	Length (Km)
1	Raghunathpur Branch	Branch	12.3
2	Parwaha Branch	Branch	22
3	Khajuri Branch	Branch	15.5
4	Mahnathpur branch	Branch	12.9
1	Portaha labatoli east minor 1	Minor	1.32
2	Portaha labatoli east minor 2	Minor	0.83
3	Chitaha east minor 1	Minor	0.12
4	Chitaha east minor 2	Minor	1.42
5	Targachi east minor	Minor	0.78
6	Barmajhiya chowk east minor	Minor	0.58
7	Barmajhiya east minor 1	Minor	0.42
8	Jitan west minor	Minor	0.9
9	Barmajhiya east minor 2	Minor	2.45
10	Dumariya east minor 1	Minor	0.68
11	Dumariya east minor 2	Minor	2.86
12	Dumariya east minor ka	Minor	1.33
13	Sabela west minor 1	Minor	3.33
14	Sabela west minor 2	Minor	4.15
15	Kothiya east minor	Minor	2.11
16	Kothiya west minor ka	Minor	0.03
17	Barkurwa east minor	Minor	1.48
18	Bagwan east minor	Minor	3.41
19	Bagwan east minor ka	Minor	1.73
20	Thilla west minor 1	Minor	0.04
21	Kulwamod east minor	Minor	1.39
22	Thilla west minor 2	Minor	0.05
23	Thilla west minor 3	Minor	0.72
24	Kharihani west minor	Minor	2.64

3.3 Command Area Distribution

3.3.1 Eastern Canal system

The command area of the East Main Canal system, considering both branch and minor canals, covers a Gross Command Area (GCA) of 21448 hectares and a Net Command Area (NCA) of 17273 hectares. For the branch canals, the GCA is 15155 hectares, with an NCA of 11977 hectares. The minor canals collectively contribute a GCA of 6293 hectares and an NCA of 5296 hectares.

Table 12 : Command area distribution in EMC

Canal Name	Canal Type	Builtup Area (Ha)	Drainage Area (Ha)	Forest (Ha)	Irrigable Area (Ha)	Grand Total (Ha)
Aaurahi BC	Branch Canal	240	19	91	1653	2003
Bokraha BC	Branch Canal	170	6	76	778	1029
Kalyanpur BC	Branch Canal	117	14	95	1071	1297
Lagdi BC	Branch Canal	416	81	398	3630	4524
Majhau BC	Branch Canal	177	9	130	1316	1632
Sikron	Branch Canal	92	4	38	743	876
Siraha BC	Branch Canal	471	30	247	1973	2721
Sukhthor BC	Branch Canal	150	9	102	812	1073
Arang	Minor Canal	35	4	20	523	582
Arnama	Minor Canal	12	2	19	136	168
Bandipur 1 ka	Minor Canal	20	12	22	547	600
Bandipur 1 kha	Minor Canal	2		0.2	28	30
Bandipur nursery minor	Minor Canal	3		5	61	69
Bandipur school minor	Minor Canal	4		18	38	60
Barhampuri	Minor Canal	10	1	16	86	112
Bhalwahi	Minor Canal	23		13	111	148
Birta	Minor Canal			5	39	44
Bulkiya left	Minor Canal	1		1	49	51
Bulkiya right	Minor Canal			7	121	128
Dakaha minor	Minor Canal	21		13	149	184
Dumri	Minor Canal	24	1	20	168	213
Gauripur	Minor Canal	10	3	13	116	142
Gautari	Minor Canal	29	4	9	210	251
Jamuwa	Minor Canal	24		16	129	169
Kalyanpur minor	Minor Canal	32	1	9	96	137

Canal Name	Canal Type	Builtup Area (Ha)	Drainage Area (Ha)	Forest (Ha)	Irrigable Area (Ha)	Grand Total (Ha)
Karjana 1	Minor Canal	8		1	27	36
Karjanha 2	Minor Canal	30	3	6	93	132
Larhaiya	Minor Canal	19	10	20	368	417
Larkana Hanumanagar	Minor Canal	72	9	31	579	692
Laxmipur	Minor Canal	54	1	34	336	425
Malhaniya pirari	Minor Canal	9		13	91	113
Manharwa	Minor Canal	40	6	11	228	285
Radhanagar	Minor Canal			2	38	40
Rajokhari minor	Minor Canal	13	3	14	191	221
Ramna minor	Minor Canal			1	37	38
Rampur	Minor Canal	4	1	9	312	326
Shiv mandir left	Minor Canal	13			18	32
Shiv mandir right	Minor Canal	2			3	6
Tikapur minor	Minor Canal	5		1	76	81
Tulsipur	Minor Canal	44	1	22	294	361
Grand Total		**2394**	**233**	**1549**	**17273**	**21448**

3.3.2 Western Canal system

The command area of the West Main Canal system, encompassing both branch and minor canals along with direct outlets, spans a Gross Command Area (GCA) of 21079 hectares and a Net Command Area (NCA) of 16502 hectares. The branch canals account for a GCA of 18768 hectares and an NCA of 14631 hectares. Meanwhile, the minor canals contribute a GCA of 2311 hectares and an NCA of 1871 hectares. Addistionally, direct outlets cover a small GCA of 17 hectares, with an NCA of 13 hectares.

Table 13 : Command area distribution in WMC

Canal Name	Canal Type	Builtup Area (Ha)	Drainage Area (Ha)	Forest (Ha)	Irrigable Area (Ha)	Grand Total (Ha)
Khajuri command area	Branch Canal	629	42	847	5191	6708
Mahinathpur Branch	Branch Canal	552	159	659	4496	5866
Parwaha Branch	Branch Canal	396	33	439	3171	4039
Raghunathpur Branch	Branch Canal	195	3	184	1773	2155
Bagwan east minor	Minor Canal	22	4	19	303	348
Bagwan east minor ka	Minor Canal				74	74

Canal Name	Canal Type	Builtup Area (Ha)	Drainage Area (Ha)	Forest (Ha)	Irrigable Area (Ha)	Grand Total (Ha)
Barkurwa east minor	Minor Canal	9	1	5	102	117
Barmajhiya chowk east	Minor Canal	16		3	11	31
Barmajhiya east minor 1	Minor Canal	4		5	10	19
Barmajhiya east minor 2	Minor Canal	15		6	127	148
Chitaha east minor 1	Minor Canal	5		1	15	22
Chitaha east minor 2	Minor Canal	8		5	59	72
Dumariya east minor 1	Minor Canal	15	1	2	19	37
Dumariya east minor 2	Minor Canal	38	1	20	149	207
Dumariya east minor ka	Minor Canal	12	1	5	45	63
Jitan west minor	Minor Canal			1	9	10
Kharihani west minor	Minor Canal	17	1	17	120	155
Kothiya minor east	Minor Canal	10	1	1	84	97
Kothiya west minor ka	Minor Canal				10	10
Kulwamod east minor	Minor Canal	3	3	1	136	143
Portaha Labatoli east minor 1	Minor Canal	8		12	53	74
Portaha labatoli east minor 2	Minor Canal	12		12	60	84
Sabela west minor 1	Minor Canal	49	1	18	251	320
Sabela west minor 2	Minor Canal	16	1	9	135	161
Targachi east minor	Minor Canal	1		1	56	58
Thilla west minor 1	Minor Canal	3		5	2	10
Thilla west minor 2	Minor Canal	1		1	10	12
Thilla west minor 3	Minor Canal	6	1	2	31	40
Grand Total		**2043**	**252**	**2283**	**16502**	**21079**

3.4 WUA Details

The Water Users Association (WUA) structure of the Kamala Irrigation Project is organized through two main canal committees: the Eastern Main Canal Committee and the Western Main Canal Committee. The Eastern Main Canal Committee supervises eight branch committees, namely Bokraha, Sikron, Maihau, Kalyanpur, Sukhthor, Siraha, Lagdi, and Aurahi along with sub-branch committees.

Similarly, the Western Main Canal Committee has four branch committees. They are: Raghunathpur, Parwaha, Khajuri, and Mainathpur and 57 sub-branch committees. Meetings are held on a monthly basis to discuss irrigation management, water allocation, and operational issues. Water distribution is coordinated through meetings between the branch committees and the project office to ensure equitable supply among users. Conflicts related to water use and allocation are resolved through joint discussions involving the irrigation office, farmers, and branch committee representatives. The major operational challenges faced by the system include disputes over water sharing among users and siltation in the canal network, which affects the efficient delivery of irrigation water. The detailed WUA information is included in **Annex I** of this report.

3.5 Rainfall Data

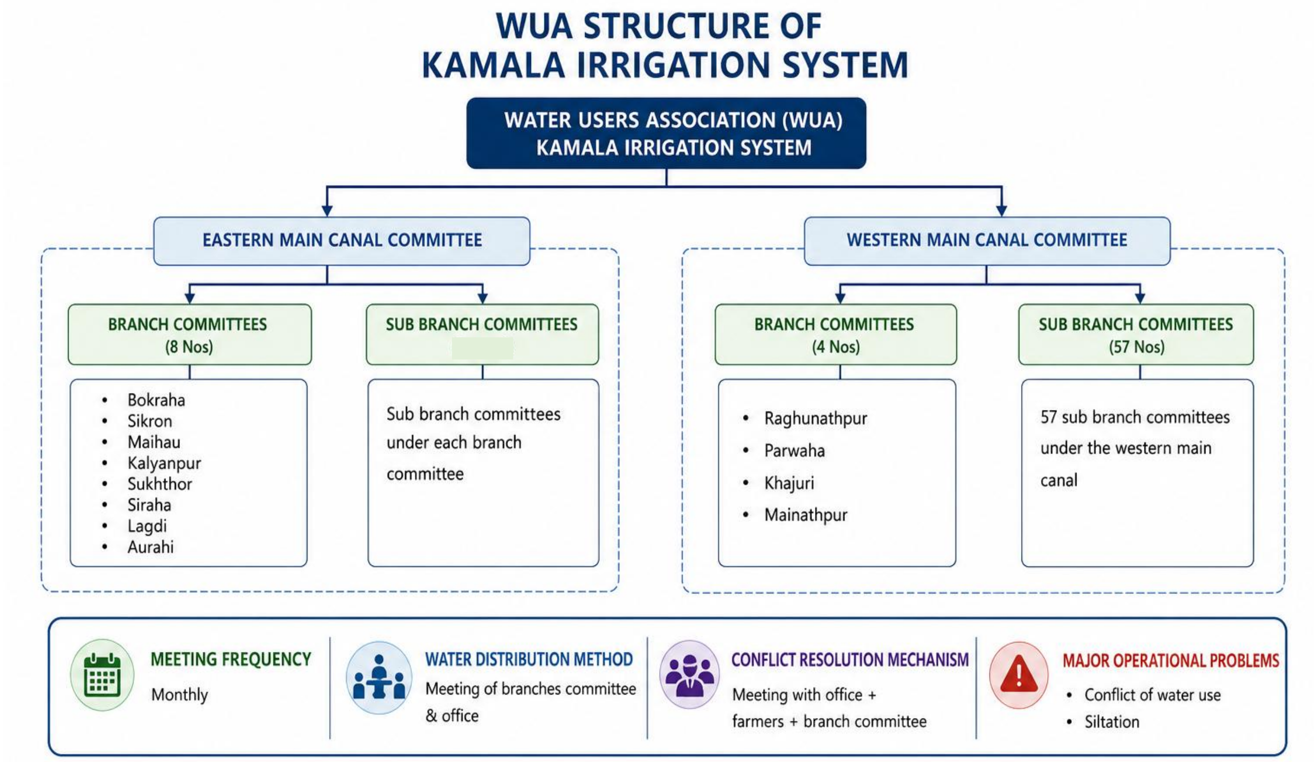


Figure 10 : WUA structure of Kamala irrigation system

The rainfall data was obtained using CLIMWAT 2.0 for Cropwat software, and yearly comparisons of total rainfall and effective rainfall were calculated using the USDA Soil Conservation Service method. The region has a significant monsoon climate, characterized by a dry period from January to May and again from October to December, followed by a heavy monsoon season spanning June to September. In the dry months, precipitation is very low, ranging from 1mm in December up to 88mm in May. However, the monsoon season brings dramatically increased precipitation, with July and August recording the highest rainfall amounts at 351mm and 354mm respectively for total rainfall.

Effective rainfall exhibits the same seasonal fluctuations but at much reduced levels, with its highest points of 160.1mm and 160.4mm occurring in July and August. This difference reflects the USDA method’s calculation of rainfall that is actually available for plant use after accounting for factors such as runoff, evaporation, and soil infiltration characteristics. The annual totals reveal that total rainfall amounts to 1,357mm while effective rainfall measures 815mm, indicating that approximately 60% of the total rainfall is effectively available for agricultural purposes in this region. The following chart is prepared based on the observed data.

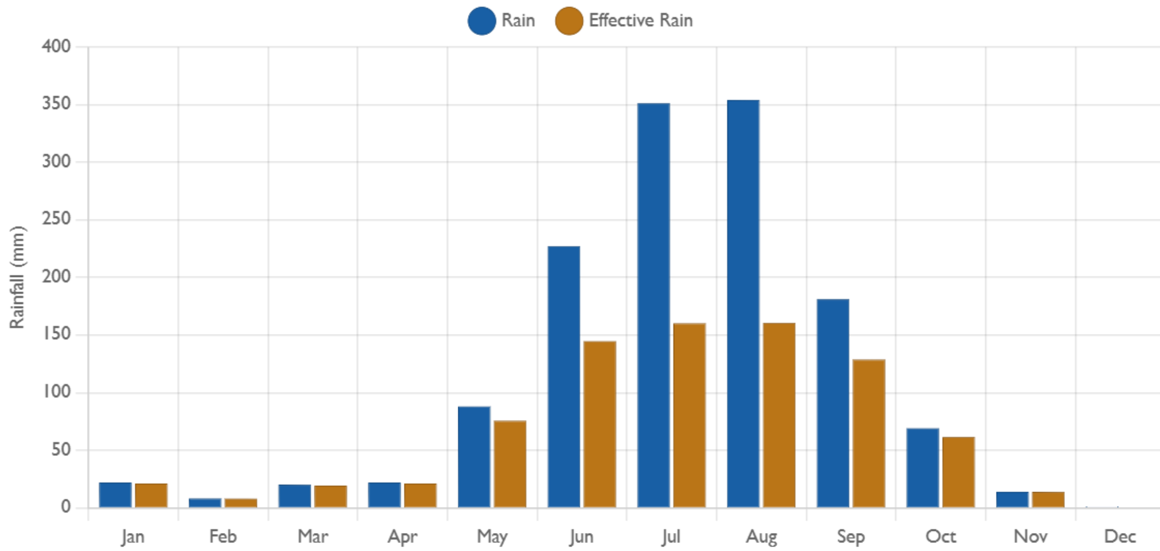


Figure 11 : Rainfall and Effective rainfall charts of the study area

3.6 Evapotranspiration Data

The Kamala Irrigation Project's command area experiences a tropical monsoon climate, according to CROPWAT 8.0 data, with noticeable changes in temperature and rainfall throughout the year. Temperatures in the region are fairly stable all year. The lowest temperatures are between 8.5°C (January) and 25.3°C (July), and the shighest temperatures hit 34.7°C in April and May, then drop. The average humidity is 73%, with peaks in September (87%) and lows in March (61%).

Wind speed fluctuates significantly; it peaks in April and May at 147 km/day and is gentler in December and January, averaging 78 km/day. Solar radiation shows a notable seasonal pattern, peaking in April (23.0 MJ/m²/day) and declining to minimum values in December and January (14.2-14.3 MJ/m²/day), directly correlating with evapotranspiration rates which follow the same trend, reaching maximum values of 5.60 mm/day in April and minimum values of 2.11 mm/day in January and December.

Table 14 : Climate data output

Month	Min Temp(°C)	Max Temp(°C)	Humidity (%)	Wind km/day	Sunshine (Hrs)	Rad MJ/m²/day	Eto (mm/day)	
Jan	8.5	22.5	72	86	7.7	14.3	2.11	
Feb	10.2	25.3	68	104	8.6	17.5	2.88	
Mar	14	30.7	61	121	8.9	20.5	4.14	
Apr	20.3	34.7	54	147	9.3	23	5.6	

Month	Min Temp(°C)	Max Temp(°C)	Humidity (%)	Wind km/day	Sunshine (Hrs)	Rad MJ/m²/day	Eto (mm/day)
May	23.7	34.4	64	147	8.4	22.5	5.58
Jun	25.2	33.6	76	130	5.5	18.3	4.51
Jul	25.3	32.3	83	121	5	17.4	4.05
Aug	24.8	32	85	104	4.8	16.6	3.75
Sep	23.9	31.4	87	95	5.2	15.8	3.46
Oct	21.6	31	75	86	7.2	16.4	3.51
Nov	14.1	28.7	73	78	8.3	15.4	2.84
Dec	9	24.3	79	78	8.2	14.2	2.11
Avg	18.4	30.1	73	108	7.3	17.7	3.71

(Source: CROPWAT 8.0 Output)

April and May are the most active climate months, demanding careful water management for irrigation, whereas December to February's cooler, drier weather means less water is needed for agriculture.

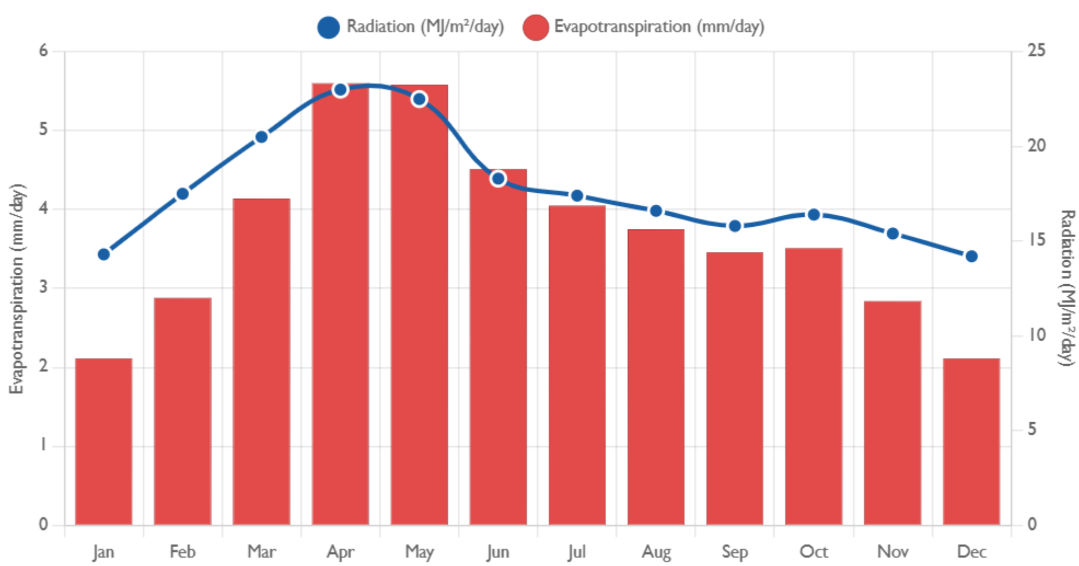


Figure 12 : Evapotranspiration (ET_0 vs Radiation) curve

(Source: CROPWAT 8.0 Output)

3.7 Land Use Distribution

The land use distribution in both command areas is predominantly covered with croplands, denoted by either Irrigable area. Additionally, ponds and rivers are categorized as drainage, alongside built-up areas.

In Kamala Eastern system, Irrigated land makes up 80.5% (17273 ha) of the total area, representing the majority.

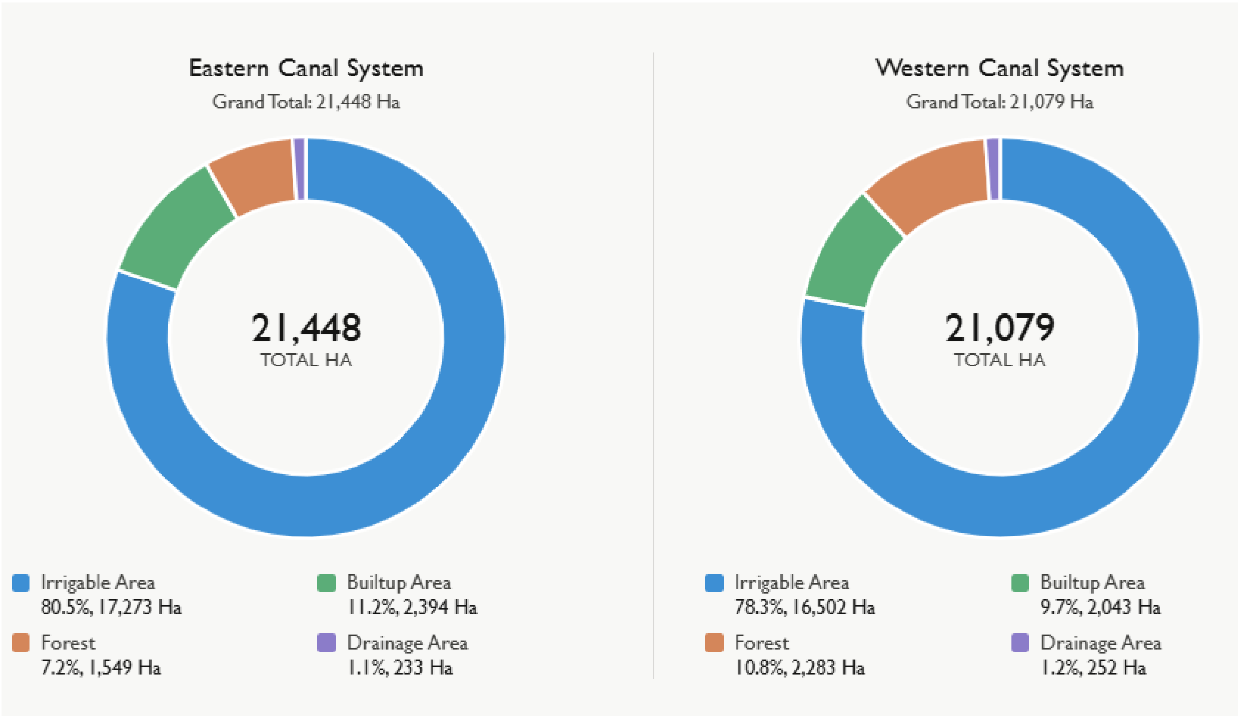


Figure 13 : Land Use Classification of the study area

(Source: Field work, 2026)

Likewise, the Kamala Western System encompasses 21079 hectares. Irrigable land makes up the largest portion of the total coverage at 78.3% (16502 ha). 10.8% (2283 ha) of the area is Forest, and Builtup area comprises 9.7% (2043 ha). Lastly, drainage areas form the smallest portion at 1.2% (252 ha).

3.8 Soil Characteristics

The command area of Kamala irrigation project is the terai region of the Kamala Basin that is characterized by predominantly flat topography and fertile alluvial floodplain soils, making it the command area for irrigated agriculture and human settlement. The soils are mainly loam and sandy loam with moderate to good water holding capacity, which supports the cultivation of major crops such as rice, wheat, and maize. Due to favorable soil texture and moisture retention properties, the Terai plays a vital role in the basin's agricultural productivity and irrigation development.

3.9 Percolation Losses

The percolation and seepage losses depend on the type of soil. They will be low in very heavy, well-puddled clay soils and high in the case of sandy soils. The percolation and seepage losses vary between 4 and 8 mm/day. FAO (1986) has suggested percolation losses value as follows:

For heavy clay: PERC = 4 mm/day
For sandy soils: PERC = 8 mm/day
On average: PERC = 6 mm/day

3.10 Conveyance and Irrigation Efficiency

In Kamala Irrigation Project, in both Eastern and Western system, irrigation water first passes through the main canal and then diverted to respective branch canal and minor canal; which is then finally applied to the agricultural field. Since the efficiencies of the system are not calibrated or documented, the main canal, branch canal, minor canal, and field application efficiencies are taken as 80%, 70%, 70%, and 70% respectively.

As efficiency decreases due to seepage, evaporation, leakage, weed growth, and improper water application in the field, this affect irrigation scheduling because more water must be supplied at the source to meet crop demand at the field level. Lower efficiency can also reduce crop production due to inadequate water availability. Improving canal maintenance, gate operation, and field irrigation methods can help increase efficiency and improve overall agricultural productivity.

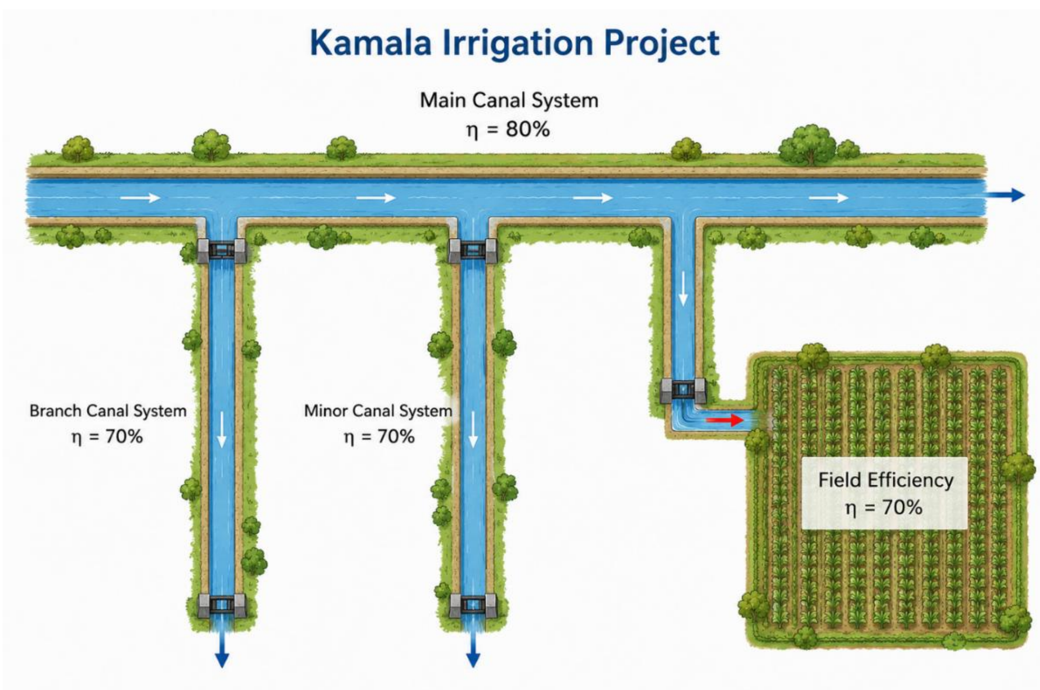


Figure 14 : Overall efficiency of the system

3.11 Cropping Pattern and Intensity

The cropping pattern within the command areas of both the Eastern Main Canal (EMC) and Western Main Canal (WMC) is dominated by rice cultivation during the monsoon season. Rice occupies approximately 90% of the cultivated area in both canal systems and is generally planted around 15 June and harvested by 12 October. During the winter season, wheat is the second major crop, covering about 65% of the cultivated area in the EMC and 60% in the WMC, with planting beginning on 1 December and harvesting around 9 April. Pulses like masur and rahar are cultivated on a smaller scale, accounting for approximately 5% of the cropped area in the EMC and 10% in the WMC, with planting typically commencing on 1 November and harvesting by 18 February. Similarly, maize occupies about 5% of the cultivated area in the EMC and 10% in the WMC, being planted around 1 December and harvested by 4 April. This cropping pattern reflects the predominance of rice-based agriculture in the project area, followed by winter crops that depend on the availability of irrigation water during the dry season.

Table 15 : Cropping pattern of EMC and WMC

S.N.	Name	Crops	Planting Date	Harvesting Date	Cropping Percentage
1	EMC	Rice	15-Jun	12-Oct	90%
		Wheat	1-Dec	9-Apr	65%
		Pulses	1-Nov	18-Feb	5%
		Maize	1-Dec	4-Apr	5%
2	WMC	Rice	15-Jun	12-Oct	90%
		Wheat	1-Dec	9-Apr	60%
		Pulses	1-Nov	18-Feb	10%
		Maize	1-Dec	4-Apr	10%

(Source: Field work, 2026)

3.12 Year-round Operational Status

Water measurement was carried out on a daily basis in both the Eastern Main Canal (EMC) and Western Main Canal (WMC) using telemetry sensor devices installed at various locations, including the intake points of both canal systems. For the Eastern Main Canal, flow data were obtained from the Kamala Irrigation Management Office, Dhanusha. During FY 2082/83, the canal remained operational for the last 15 days of Ashadh, throughout Shrawan, and throughout Bhadra except for a few days when csanal operation was closed due to flooding. In Ashoj, the canal was operated for 5 days, while in Kartik it remained in operation for the first 11 days of the month. The canal was completely closed during Mangsir. In Poush, it was reopened and operated during the last 17 days of the month, and in Magh it remained operational for the first 12 days. The canal was fully closed throughout Falgun, Chaitra, Baisakh, and Jestha.

Similarly, for the Western Main Canal, the canal was operational for the last 15 days of Ashadh, throughout Shrawan, and throughout Bhadra except for a few days when operation was halted due to flood events. In Ashoj, the canal was operated for 6 days, and in Kartik it remained operational during the first 11 days of the month. The canal was completely closed throughout Mangsir. In Poush, it was operated during the last 17 days of the month, while in Magh it remained open for the first 12 days. The canal was fully closed during Falgun, Chaitra, Baisakh, and Jestha.

Figure 15 : Telemetry and staff reading for discharge measurement

(Source: Fieldwork, 2026)

3.13 Available discharge at main canal

The available discharge at the main canal of both EMC and WMC was monitored continuously using telemetry sensor devices installed at the canal intake and other key locations along the Kamala irrigation system. Daily discharge data from the month shrawan to magh were obtained from the Kamala Irrigation Management Office, Dhanusha, and analyzed to assess the temporal variation in water availability during FY 2082/83. The discharge pattern was directly governed by the canal operation schedule, with flows recorded during periods when the canal was in operation and negligible or no discharge during closure periods. Higher discharges were generally observed during the monsoon months when water availability in Kamala river was abundant, while reduced flows or interruptions occurred during flood events and canal closure periods. The daily discharge versus month graph presented below illustrates the variation in available canal discharge throughout the fiscal year and provides an overview of the seasonal distribution of irrigation water supply in the system.

Table 16 : Discharge values at EMC and WMC

Annual values of	Kamala EMC	Kamala WMC
Mean discharge	2.93 m^3/s	3.06 m^3/s
Peak discharge	12.88 m^3/s	14.28 m^3/s
Total zero-flow days	121 (54%)	131 (58%)

Annual values of	Kamala EMC	Kamala WMC
Total discharge volume (sum of daily Q)	656.6	684.9

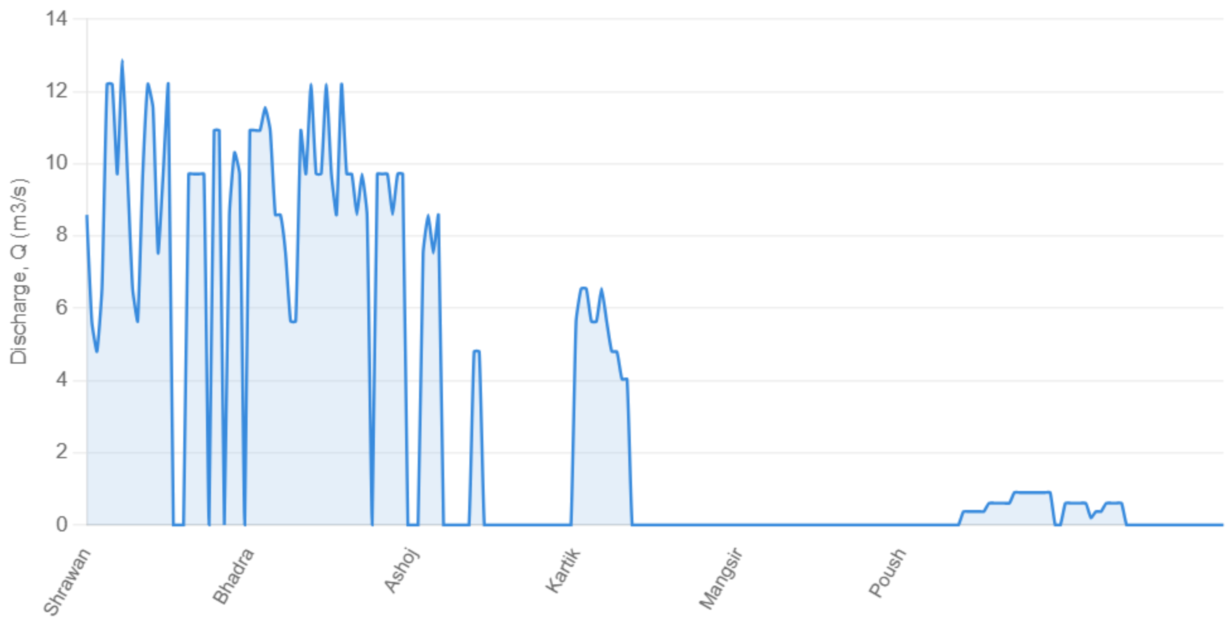


Figure 16 : EMC yearly discharge data

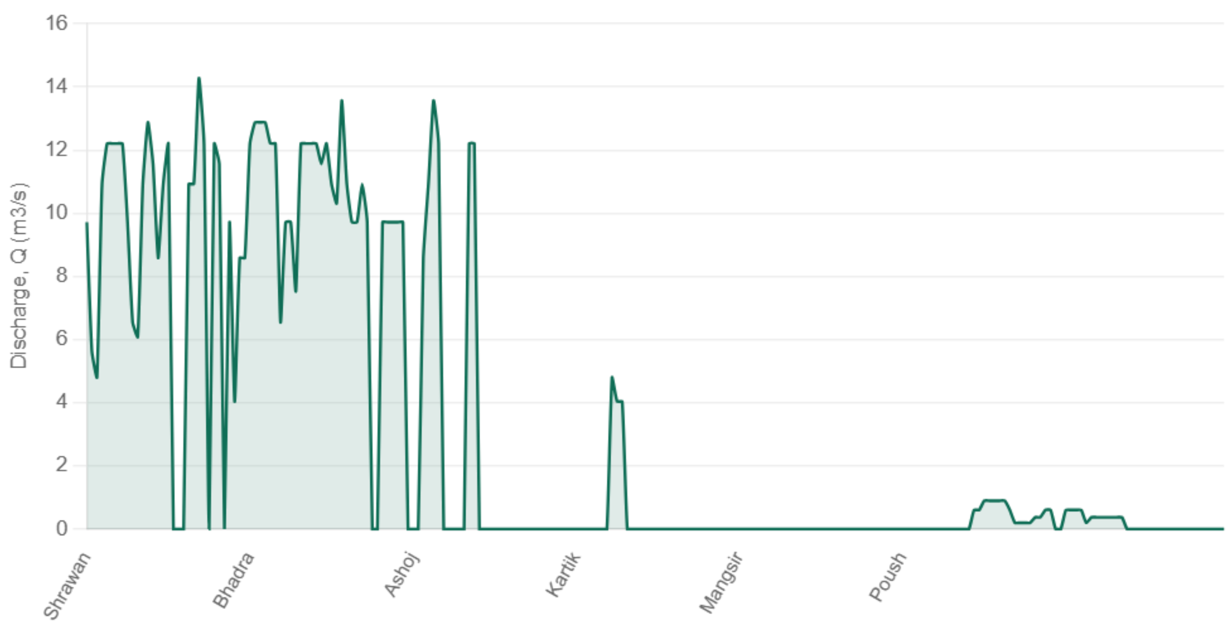


Figure 17 : WMC yearly discharge data

(Source: Modified form discharge data, Kamala irrigation management office, portaha)

4 Irrigation Water Requirements and Scheduling

4.1 Water Resources Availability

The discharge measurements were available from the Chisapani hydrological station (26.421°N, 86.175°E) in Dhanusha district for the periods 1956–1970 and 2000–2004 for kamala river. According to the Koshi Basin Master Plan Study (JICA, 1985), the mean monthly flow of the Kamala River recorded at Chisapani station (Station 598) during 1956–1970 was 44.70 m^3/s. The flow records indicate significant seasonal variation, with extreme fluctuations in monthly streamflow primarily influenced by rainfall intensity and quantity. During the dry season, rivers originating from the Chure region often experience very low discharge or become non-perennial, whereas during the monsoon season they carry high volumes of water and sediment. River flows respond rapidly to rainfall events, with water levels rising and declining within a few hours due to the high infiltration and permeability characteristics of the river corridor.

Table 17: Monthly Flow pattern in Kamala River

Months	Min flow (m^3/s)	Max flow (m^3/s)	Mean flow (m^3/s)
Jan	4.0	11.6	6.8
Feb	1.0	6.8	4.4
Mar	2.0	6.8	4.4
Apr	2.0	4.4	2.0
May	2.0	25.2	7.6
Jun	17.0	84.4	43.6
Jul	31.0	318.0	130.8
Aug	53.0	395.6	155.6
Sep	38.0	189.2	102.8
Oct	20.0	73.2	46.0
Nov	4.0	32.6	17.2
Dec	8.0	15.3	9.2

(Source: WECS Report on Water Resources Development Strategy for the Kamala River Basin, Nepal, 2021)

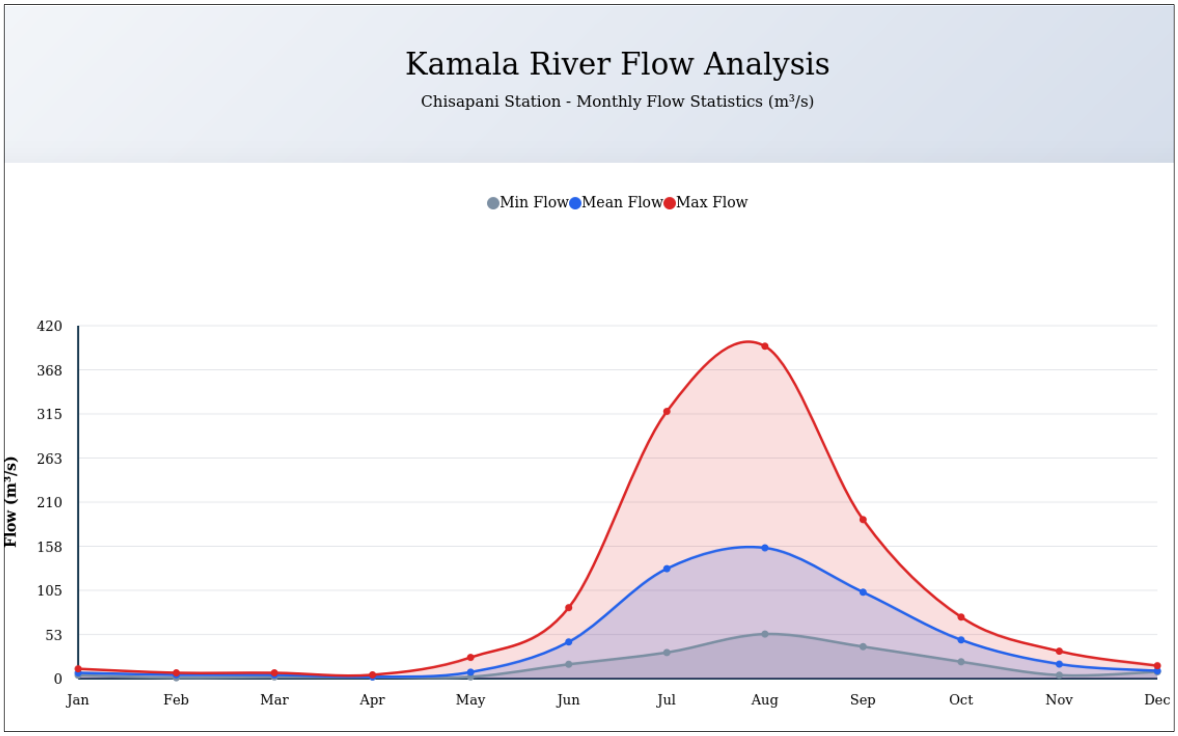


Figure 18 : Monthly hydrograph of Kamala Rivers

Therefore, the Kamala River exhibits pronounced seasonal variation, with peak flows occurring during the monsoon months (June-September). August represents the highest mean discharge at 155.6 m^3/s, while the dry season (December-May) shows minimal flows averaging 4.0 m^3/s or less.

4.2 Water Balance and Sufficiency Analysis

4.2.1 For EMC

The water balance analysis for the Eastern Main Canal (EMC) of the Kamala Irrigation System was carried out using crop water requirements estimated through CROPWAT and available discharge measured daily in the main canal for the FY year 2082/83. The analysis indicates significant variation between irrigation water demand and canal water availability throughout the year especially for winter crops. The peak crop water requirement was estimated at 8,291 l/s during June, due to the extensive cultivation of rice over approximately 80% of the command area. Water deficits were observed during January, February, March, April, June, October, and December, with the highest deficit occurring in February. These deficits indicate that the available canal discharge during these months was insufficient to meet the crop water requirements, potentially affecting crop growth and yields if supplemental irrigation sources are not available. In contrast, water surpluses were recorded during May, July, August, September, and November.

To address the water shortages observed during deficit months, several measures may be considered. Improvement of canal operation and maintenance practices, including reduction of conveyance losses through canal lining and timely desiltation, can enhance water delivery efficiency. Adoption of rotational water distribution schedules and improved on farm water management practices can further optimize water use during critical periods. Additionally, conjunctive use of groundwater and surface water, where feasible, can overcome irrigation deficits during peak demand months. Consideration may also be given to adjusting cropping patterns, promoting less water intensive crops during water scarce periods. The water balance calculations is shown in the table below.

4.2.2 For WMC

The water balance analysis for the Western Main Canal (WMC) of the Kamala Irrigation System was also carried out using crop water requirements estimated through CROPWAT. These results also indicates a considerable variation between irrigation demand and available canal discharge throughout the year. The peak crop water requirement was estimated at 7,921 l/s during June, corresponding to the period of maximum rice cultivation within the command area. Water deficits were observed during January, February, March, April, May, June, October, and December indicating that canal supplies during these periods were insufficient to satisfy the estimated crop water requirements. Conversely, substantial water surpluses were observed during the monsoon and post-monsoon months. The highest surplus occurred in July, followed by August and September when canal flows significantly exceeded irrigation demands due to reduced irrigation requirements. To address the water shortages observed during deficit months, same measures as discussed earlier may be considered. The water balance calulations is shown in the table below.

Table 18 : Water Balance calculations for KIS, Eastern Main canal (EMC)

Particulars	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Precipitation deficit												
1. Rice	0	0	0	0	1.3	180.4	0	0	2.2	39.1	0	0
2. Spring Wheat	29.6	79.1	92.6	13.5	0	0	0	0	0	0	0	19.3
3. Maize	0	0	26.6	126.8	112.9	7.9	0	0	0	0	0	0
Net scheme irr.req.												
in mm/day	0.6	1.7	1.9	0.9	0.6	5.1	0	0	0.1	1.1	0	0.4
in mm/month	17.8	47.5	59.5	27.2	18	154.5	0	0	1.8	33.2	0	11.6
in l/s/h	0.07	0.2	0.22	0.1	0.07	0.6	0	0	0.01	0.12	0	0.04
Irrigated area (% of total area)	45	45	10	10	10	80	0	0	80	80	0	45
Irr.req. for actual area (l/s/h)	0.11	0.33	0.3	0.14	0.07	0.6	0	0	0.01	0.15	0	0.07
Cropped Area (Ha)	7773	7773	1727	1727	1727	13818	0	0	13818	13818	0	7773
Total water requirement, lps	544	1555	380	173	121	8291	0	0	138	1658	0	311
Monthly canal flow, lps	0	0	0	0	2055	6167	8940	5525	1694	997	308	308
Water balance, lps	-544	-1555	-380	-173	1934	-2124	8940	5525	1556	-661	308	-3
Peak CWR	8291											

Table 19 : Water Balance calculations for KIS, Western Main canal (WMC)

Particulars	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Precipitation deficit												
1. Rice	0	0	0	0	1.3	180.4	0	0	2.2	39.1	0	0
2. Spring Wheat	29.6	79.1	92.6	13.5	0	0	0	0	0	0	0	19.3
3. Maize	0	0	26.6	126.8	112.9	7.9	0	0	0	0	0	0
Net scheme irr.req.												
in mm/day	0.6	1.7	1.9	0.9	0.6	5.1	0	0	0.1	1.1	0	0.4
in mm/month	17.8	47.5	59.5	27.2	18	154.5	0	0	1.8	33.2	0	11.6
in l/s/h	0.07	0.2	0.22	0.1	0.07	0.6	0	0	0.01	0.12	0	0.04
Irrigated area (% of total area)	45	45	10	10	10	80	0	0	80	80	0	45
Irr.req. for actual area (l/s/h)	0.11	0.33	0.3	0.14	0.07	0.6	0	0	0.01	0.15	0	0.07
Cropped Area (Ha)	7426	7426	1650	1650	1650	13202	0	0	13202	13202	0	7426
Total water requirement, lps	520	1485	363	165	116	7921	0	0	132	1584	0	297
Monthly canal flow, lps	0	0	0	0	0	6808	9800	6423	1376	214	238	238
Water balance, lps	-520	-1485	-363	-165	-116	-1113	9800	6423	1244	-1370	238	-59
Peak CWR	7921											

4.3 Crop Water Requirements for Major Crops

The major crops grown in both canal systems of kamala irrigation project are classified on the basis of larger cultivation area. Thus, paddy, wheat and maize are categorized ass major crops in this region and CWR calculations are done using Cropwat software.

1. Rice

Rice is the major cultivated crop in the study area, requiring a reliable and timely supply of irrigation water throughout its growing period. Due to its relatively highwater demand and sensitivity to watesr stress during critical growth stages such as transplanting, tillering, flowering, and grain filling, proper irrigation scheduling is essential to achieve optimum yield and efficient water use. Therefore, preparation of an irrigation schedule for rice is necessary to determine the timing and quantity of water application according to crop growth stages, climatic conditions, and field requirements. The different stages are shown in given figure.

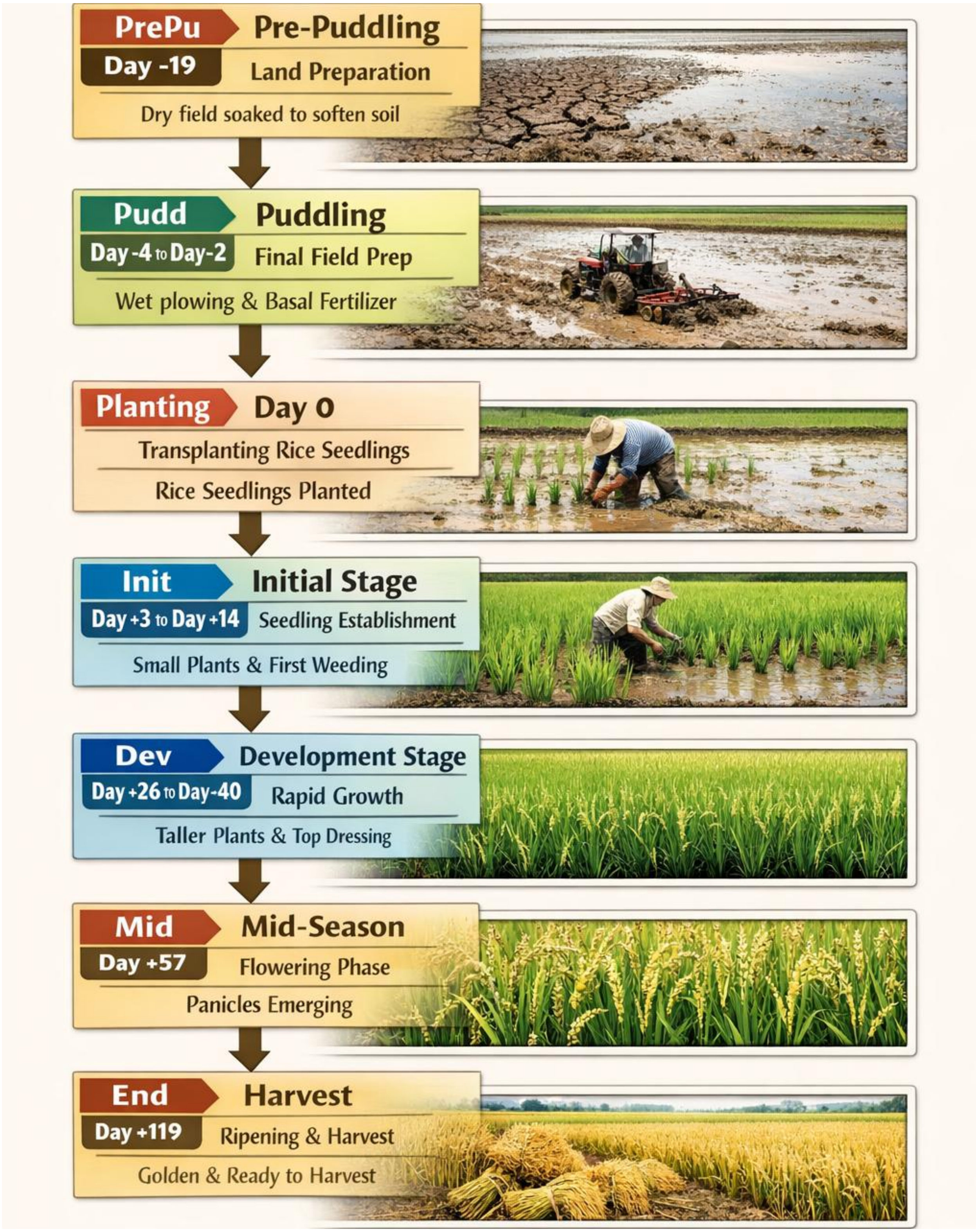


Figure 19 : Different stages in Rice Cultivation

The crop water requirement analysis for rice, calculated using the CROPWAT software, showed that the rice crop has varying water demand throughout its growth stages. The crop coefficient (Kc) ranges from about 0.99 to 1.20, with higher values during the nursery and mid-season stages, indicating greater evapotranspiration and water requirement during these periods. The total crop evapotranspiration (ETc) for the growing season was estimated to be about 589.7 mm, while effective rainfall contributed approximately 654.1 mm. As a result, the net irrigation requirement was relatively low, 222.9 mm, with irrigation mainly required during the early establishment and late growth stages when rainfall was insufficient to meet crop water demand of rice.

Table 20 : Irrigation required for different stages for Rice crop

Month	Decade	Stage	Kc	Etc (mm/day)	Etc (mm/dec)	Eff rain (mm/dec)	Irr. Req. (mm/dec)
May	3	Nurs	1.2	0.63	1.3	5.9	1.3
Jun	1	Nurs/LPr	1.17	1.5	15	42.2	90.8
Jun	2	Nurs/LPr	1.06	4.8	48	50.7	0
Jun	3	Init	1.07	4.67	46.7	51.6	89.6
Jul	1	Init	1.1	4.62	46.2	52.1	0
Jul	2	Deve	1.1	4.46	44.6	54.1	0
Jul	3	Deve	1.1	4.36	48	53.9	0
Aug	1	Deve	1.11	4.28	42.8	54.4	0
Aug	2	Mid	1.12	4.19	41.9	55	0
Aug	3	Mid	1.12	4.08	44.9	50.9	0
Sep	1	Mid	1.12	3.97	39.7	47.5	0
Sep	2	Mid	1.12	3.86	38.6	44.5	0
Sep	3	Late	1.11	3.87	38.7	36.5	2.2
Oct	1	Late	1.08	3.75	37.5	27.3	10.2
Oct	2	Late	1.03	3.61	36.1	19.6	16.6
Oct	3	Late	0.99	3.27	19.6	7.9	12.3
Total					**589.7**	**654.1**	**222.9**

(Source: CROPWAT 8.0 Output)

2. Wheat

Wheat is other major winter season crop cultivated here and plays an important role in local food security and agricultural production. It is widely grown after the monsoon rice harvest and requires controlled irrigation during the dry season due to limited rainfall availability. Proper irrigation scheduling for wheat is therefore essential to ensure optimum crop growth, improve water use efficiency, and achieve higher yields.

The crop water requirement analysis for wheat, estimated using the CROPWAT software, indicates that the crop water demand gradually increases from the initial stage to the mid-season stage and decreases towards maturity. The crop coefficient (Kc) varies from about 0.30 during the initial stage to a peak of about 1.15 during the middle stage, indicating higher evapotranspiration and water requirement during active growth periods. The total seasonal crop evapotranspiration (ETc) was estimated to be approximately 306 mm, while effective rainfall contributed about 54.0 mm during the growing period. Consequently, the net irrigation requirement was around 251.4 mm, with supplemental irrigation mainly required during the middle stages when rainfall was insufficient to satisfy crop water demand.

Table 21 : Irrigation required for different stages for wheat crop

Month	Decade	Stage	Kc	Etc (mm/day)	Etc (mm/dec)	Eff rain (mm/dec)	Irr. Req. (mm/dec)
Dec	1	Init	0.3	0.71	7.1	0.4	6.6
Dec	2	Init	0.3	0.63	6.3	0	6.3
Dec	3	Deve	0.3	0.64	7	0.7	6.3
Jan	1	Deve	0.48	1.02	10.2	5.9	4.3
Jan	2	Deve	0.77	1.61	16.1	8.6	7.5
Jan	3	Mid	1.06	2.49	27.4	6.6	20.8
Feb	1	Mid	1.15	3.01	30.1	3.3	26.7
Feb	2	Mid	1.15	3.3	33	1.5	31.6
Feb	3	Mid	1.15	3.79	30.3	3.1	27.1
Mar	1	Mid	1.15	4.27	42.7	5.4	37.2
Mar	2	Late	0.99	4.11	41.1	6.9	34.2
Mar	3	Late	0.7	3.22	35.4	6.9	28.5
Apr	1	Late	0.41	2.14	19.3	4.7	14
Total					**306**	**54.1**	**251.4**

(Source: CROPWAT 8.0 Output)

3. Maize

Maize is another major winter season crop cultivated here. It is widely grown after the monsoon rice harvest and requires controlled irrigation during the dry season due to limited rainfall availability.

The crop coefficient (Kc) varies from about 0.30 during the initial stage to a peak of about 1.1 during the mid-season stage, indicating higher evapotranspiration and water requirement during active growth periods. The total seasonal crop evapotranspiration (ETc) was estimated to be approximately 349 mm, while effective rainfall contributed about 521 mm during the growing period. Consequently, the net irrigation requirement was relatively low, totaling around 20.3 mm, with supplemental irrigation mainly required during the late growth stages when rainfall was insufficient to satisfy crop water demand.

Table 22 : Irrigation required for different stages for Maize crop

Month	Decade	Stage	Kc	Etc (mm/day)	Etc (mm/dec)	Eff rain (mm/dec)	Irr. Req. (mm/dec)
Jun	3	Init	0.3	1.31	2.6	10.3	2.6
Jul	1	Init	0.3	1.26	12.6	52.1	0
Jul	2	Deve	0.31	1.24	12.4	54.1	0
Jul	3	Deve	0.48	1.91	21	53.9	0
Aug	1	Deve	0.72	2.78	27.8	54.4	0
Aug	2	Deve	0.95	3.57	35.7	55	0
Aug	3	Mid	1.1	4.01	44.1	50.9	0
Sep	1	Mid	1.1	3.91	39.1	47.5	0
Sep	2	Mid	1.1	3.8	38	44.5	0
Sep	3	Mid	1.1	3.82	38.2	36.5	1.7
Oct	1	Late	0.99	3.44	34.4	27.3	7.1
Oct	2	Late	0.74	2.58	25.8	19.6	6.3
Oct	3	Late	0.47	1.56	17.2	14.6	2.6
Total					**349**	**520.6**	**20.3**

(Source: CROPWAT 8.0 Output)

4.4 Irrigation Depth, Frequency, and Duration

1. Rice

The irrigation depth for rice crop was estimated using the CROPWAT software and scheduled into four irrigation batches according to water availability during different stages of crop growth as shown in Table. The first irrigation shall be applied on 9th June during the pre-puddling stage (PrePu), with a net irrigation depth of 91.6 mm. The second irrigation as per CROPWAT is scheduled on 24th June during the puddling stage (Puddl), where 89.4 mm of water is required. The third irrigation shall be applied around 2nd July at the initial growth stage (Init), requiring the highest net irrigation depth of 98.2 mm.

Finally, the fourth irrigation shall be applied on 14th October during the late/end growth stage, with a net irrigation depth of 96.7 mm is desired for rice crop. The irrigation schedule is designed to optimize available water resources while meeting the water demand of the rice crop throughout its growth cycle.

Table 23 : Net irrigation depth and timings of irrigation for Rice crop

Date	Day	Stage	Rain (mm)	Ks	Percolation (mm)	Net irrigation (mm)	Loss (mm)
9-Jun	-19	PrePu	0	0.9	0	91.6	0
24-Jun	-4	Puddl	0	1	6.3	89.4	0
2-Jul	4	Init	0	1	3.1	98.2	0
14-Oct	108	End	0	1	3.1	96.7	0
26-Oct	End	End	0	1	0		

(Source: CROPWAT 8.0 Output)

2. Wheat

The irrigation schedule for the wheat crop was determined and planned according to crop water requirements and available water supply during its growing season. An irrigation application is scheduled on 1st July during the initial growth stage (Init), where no rainfall was recorded and the crop stress coefficient (Ks) was 1, indicating no water stress. A net irrigation depth of 38.4 mm is required to satisfy crop water demand, resulting in a gross irrigation depth of 54.9 mm with an estimated flow rate of 2.12 l/s/ha.

This irrigation scheduling approach ensured efficient water management and adequate moisture availability for optimum wheat crop growth and development throughout the season.

Table 24 : Net irrigation depth and timings of irrigation for Wheat crop

Date	Day	Stage	Rain (mm)	Ks	Net Irr (mm)	Gr. Irr (mm)	Flow (l/s/ha)
6-Dec	6	Init	0	1	43.2	61.7	1.19
2-Feb	64	Mid	0	1	134	191.4	0.38
20-Mar	110	End	0	1	152	217.2	0.55
9-Apr	End	End	0	1			

(Source: CROPWAT 8.0 Output)

3. Maize

The irrigation schedule for Maize crop was determined and planned according to crop water requirements and available water supply during its growing season. An irrigation application is scheduled on 1st July during the initial growth stage (Init), where no rainfall was recorded. A net irrigation depth of 37.8 mm is required to satisfy crop water demand, resulting in a gross irrigation depth of 53.9 mm with an estimated flow rate of 2.08 l/s/ha.

This irrigation scheduling approach ensured efficient water management and adequate moisture availability for optimum wheat crop growth and development throughout the season.

Table 25: Net irrigation depth and timings of irrigation for Maize crop

Date	Day	Stage	Rain	Ks	Net Irr (mm)	Gr. Irr (mm)	Flow (l/s/ha)
1-Jul	3	Init	0	1	37.8	53.9	2.08
31-Oct	End	End	0	1			

(Source: CROPWAT 8.0 Output)

4.5 Irrigation Scheduling

4.5.1 When Water is Sufficient (Continuous Flow)

When water in Kamala River and ultimately in main canal system becomes available in sufficient quantity throughout the considered irrigation season, a continuous flow irrigation schedule should be adopted. The canal system runs continuously, with all branches and minors opened all at same time and allowing farmers at head, middle and tail section to irrigate their fields whenever required. This method provides flexibility in water application and helps crops receive adequate moisture at critical growth stages.

4.5.2 When Water is Insufficient (Rotational Scheduling)

When the available water from kamala river supply becomes limited and cannot meet the demand of all users at head, middle and tail simultaneously, rotational scheduling must be implemented. Under this system, water is distributed to different areas or groups of farmers according to a timetable obtained from the developed excel sheet. This ensures equitable water distribution, reduces conflicts among users, and improves the efficient use of scarce water resources.

5 Water Allocation, Distribution and Delivery Schedules

5.1 Development of Scheduler in Excel and Dashboard

To improve the efficiency and equity of irrigation water distribution, an Excel-based water allocation and scheduling tool, accompanied by an interactive dashboard, was developed. The tool generates optimized irrigation schedules using **First-Fit Decreasing** (FFD) algorithm by calculating the required duration and timing of water delivery for each branch canal. The calculations are based on key parameters including the command area served by each branch, irrigation depth requirement (obtained from CROPWAT analysis), available canal discharge, irrigation efficiency, and the desired number of irrigation batches. Using these inputs, the system automatically determines water allocation volumes and delivery schedules with specific dates and times for each irrigation cycle. The dashboard provides a user-friendly interface for visualizing water demand, monitoring distribution plans, and supporting operational decision-making. This approach enhances water-use efficiency, promotes equitable distribution among farmers, and facilitates systematic canal operation and management.

Table 26 : Inputs and output features of developed scheduler

Level	Category	Parameter	Unit	Input / Output
System	Main Canal	Main canal flow	lps	Input
Canal	Identity	Canal name	-	Input
Canal	Hydraulics	Flow available	lps	Input
Canal	Hydraulics	Command area	ha	Input
Canal	Efficiency	Canal efficiency	%	Input
Canal	Depth	Default irrigation depth	mm	Input
Batch	Scheduling	Batch start date & time	Date/Time	Input
Batch	Depth	Per-canal depth per batch	mm	Input
Canal/Batch	Volume	Net irrigation volume	m^3	Output
Canal/Batch	Volume	Gross irrigation volume	m^3	Output
Canal/Batch	Timing	Irrigation duration	hrs / days	Output
Canal/Batch	Timing	Start date & time	Date/Time	Output
Canal/Batch	Timing	End date & time	Date/Time	Output
Canal/Batch	Efficiency	Duty of water	ha/cumec	Output
Canal/Batch	Operation	Gate status (Open / Closed)	-	Output
System	Block Grouping	Block assignment per canal	-	Output
System	Block Grouping	Number of blocks formed	Count	Output
System	Supply Check	Total demand vs. supply	lps	Output
System	Supply Check	Block cycle duration	hrs	Output

5.2 Water Delivery schedule from Main Canal to Branch Canals and Minor canals

5.2.1 For EMC

From the developed excel sheet, water delivery scheduling with exact dates and timings for every branch and minor canals was developeds for major crops rice and wheat. The water delivery for rice is shown in figure below and the full schedule for EMC is included in **Annex 2** of this report.

The EMC of Kamala Irrigation Project comprises a total of 40 canals, which are scheduled in 4 phased batches, as per CROPWAT results. In the flow chart figure, only three representative canals are shown in the flow chart, while the focus remains on the batch wise implementation schedule. The 1st Batch is scheduled to commence on 09 June 2026, followed by the 2nd Batch on 24 June 2026, the 3rd Batch on 02 July 2026 and the last batch on. These start dates indicate the phased rollout strategy adopted for canal operations within EMC, ensuring systematic distribution and management of irrigation water across the command area.

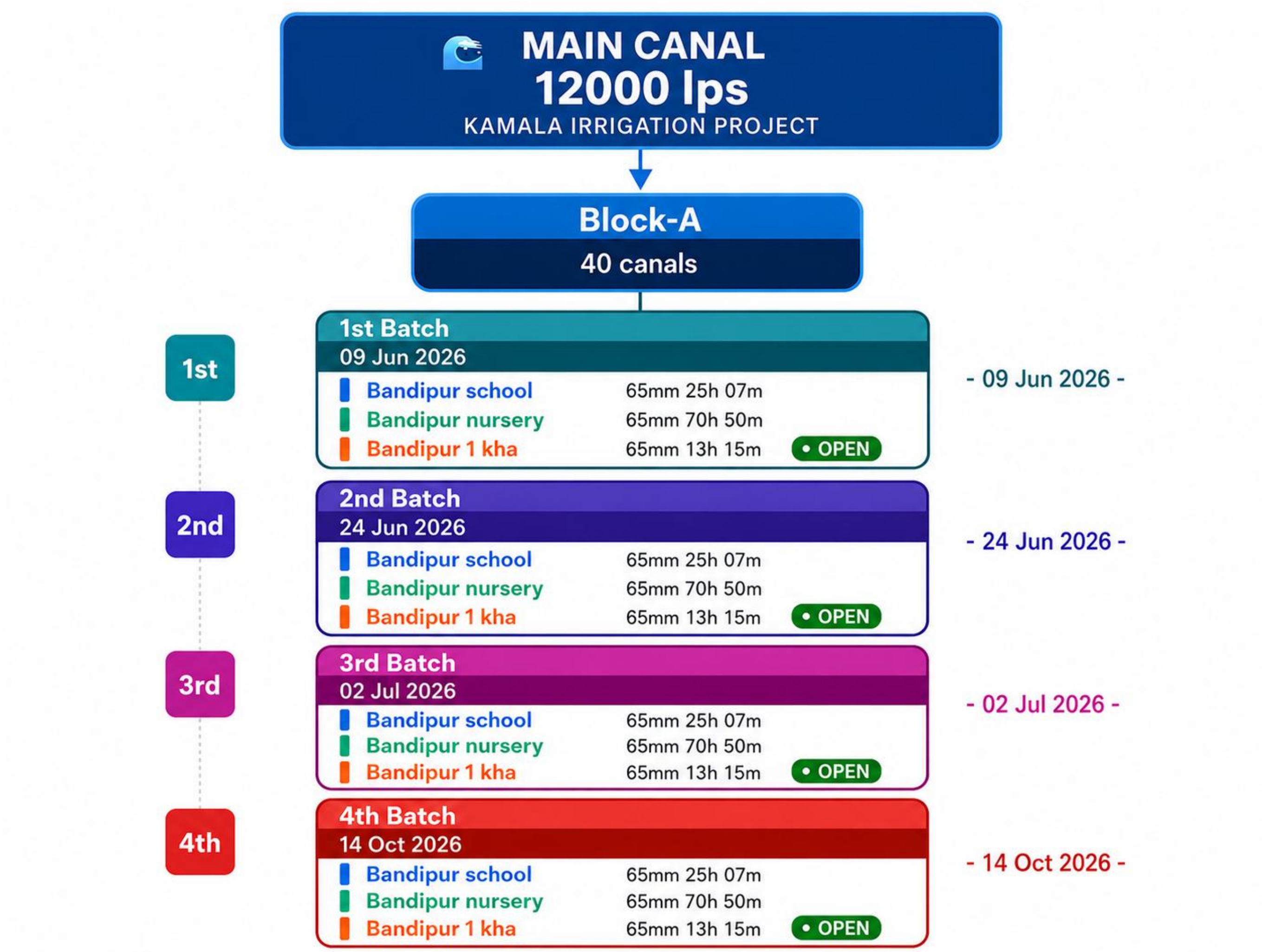


Figure 20 : Water delivery schedule in 4 batches for Rice crop in EMC

5.2.2 For WMC

Kamala WMC consists of a total of 28 canals i.e 4 branch canals and 24 minor canals along the course. The irrigation schedule is developed and planned to be brought into operation in a 4 phased manner as obtained from CROPWAT results. Due to space limitations in the flow chart, only three representative canals are displayed; however, the emphasis is on the batch wise implementation schedule. The 1st Batch is scheduled to commence on 09 June 2026, followed by the 2nd Batch on 24 June 2026, the 3rd Batch on 02 July 2026, and the 4th Batch on 14 October 2026. The full schedule of WMC is included in **Annex 3** of this report.

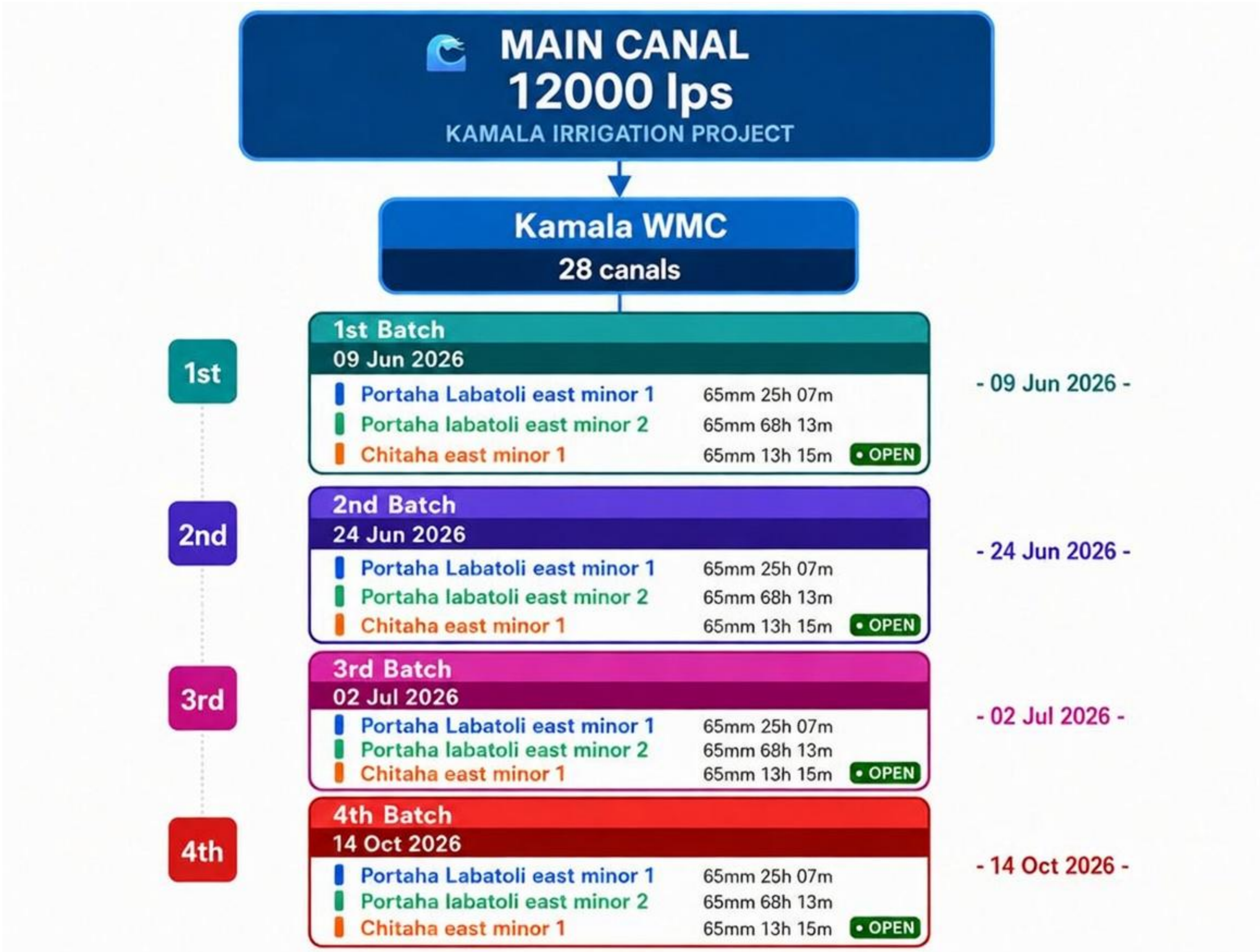


Figure 21: Water delivery schedule in 4 batches for Rice crop in WMC

6 Organization, Roles and Responsibilities

6.1 Existing Organization Structure

The Kamala Irrigation Management Office (KIMO), located in Portaha, Dhanusha, operates under the Government of Nepal and is responsible for the management, operation, and development of irrigation infrastructure within its service area. The organization is structured to ensure effective planning, implementation, and monitoring of irrigation activities while delivering quality services to the agricultural community.

The office is headed by an Office Chief, who provides overall leadership and supervises both technical and administrative functions. Supporting the Office Chief are technical personnel, including engineers and other professional staff, who are responsible for the planning, design, construction, operation, and maintenance of irrigation systems. Administrative and financial functions are managed by support staff, including an Accountant, who oversees budgeting, financial management, and record keeping.

To facilitate efficient service delivery, the office is organized into different divisions and sections, each assigned specific responsibilities. These divisions coordinate closely to implement irrigation projects, manage water resources, and provide technical assistance to farmers. The organizational framework promotes effective communication, accountability, and timely decision making in the execution of every management and operational tasks.

The major operational components of the office focus on three key areas. First, the Irrigation Projects component is responsible for planning, designing, and implementing infrastructure development activities aimed at improving irrigation facilities. Second, the Water Management component oversees the distribution, regulation, and monitoring of water resources to ensure equitable and efficient utilization. Third, the Farmer Assistance component provides support services, technical guidance, and coordination with water users and farming communities to enhance agricultural productivity.

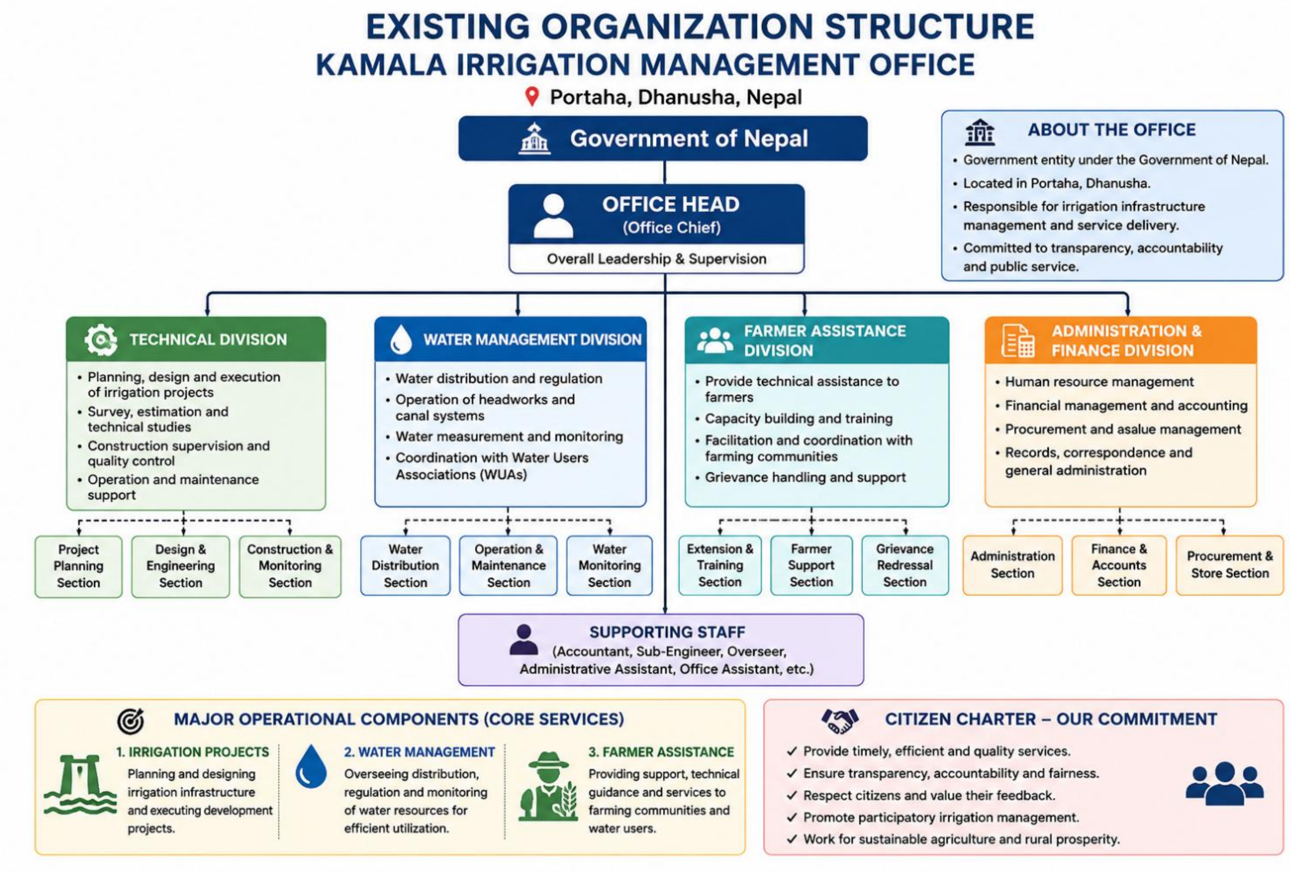


Figure 22 : Existing organization structure of Kamala irrigation management office

6.2 Clear Roles and Responsibilities

6.2.1 Main Canal Operation and Water Distribution

1. Duties of the Division Chief

- The overall responsibility for operating the main canal and distributing water from the main canal to branch canals and minor canals according to the operation plan and water distribution schedule lies with the Division Chief.
- For this purpose, the Division Chief shall assign duties and responsibilities to Engineers, Sub-engineers, and canal guards as required.
- The Division Chief shall coordinate with the Central and Regional Committees of the Water Users Association to fix the start and end dates of canal operation for each crop season and disseminate this information to the public.
- The Division Chief must inform local administration about the canal operation schedule and ensure the safety of staff and representatives involved in canal operation.
- If repair and maintenance work is required during canal operation, the Division Chief shall promptly inspect and arrange necessary repairs.

- The Division Chief shall regularly supervise and monitor the work of Engineers responsible for canal operation.

2. Duties of the Central Water Users Association Committee

- To take interest in the operation of the main canal and provide necessary support.
- To monitor daily whether water is being distributed to branch canals and minor canals according to the schedule and provide suggestions accordingly.
- To assist in monitoring water distribution and canal patrolling by assigning committee members or representatives.

3. Duties of the Engineer

- To mobilize Sub-engineers and canal guards for safe and efficient canal operation and ensure water distribution to branch canals, main canals, and direct tertiary canals according to the schedule.
- To ensure safety and systematic operation of canals and irrigation structures, including daily patrolling of the main canal.
- To generally maintain full supply level in branch and main canals, while coordinating with Block Committee representatives to regulate or stop supply as needed.
- To maintain daily records of water released from the main canal to branch and main canals, including Discharge and duration.
- To keep records of water received at the headworks and distributed downstream, and share this information with the division office and relevant committees.
- To arrange for prompt repair and maintenance of canals and irrigation structures if damaged.
- During winter and spring seasons, when water supply is low, to make appropriate arrangements to ensure equitable water distribution.
- To supervise Sub-engineers and canal guards and provide necessary instructions.
- To maintain regular communication with the Division Chief and provide updates.
- To identify sensitive locations where canal operation may be disrupted and take special precautions.

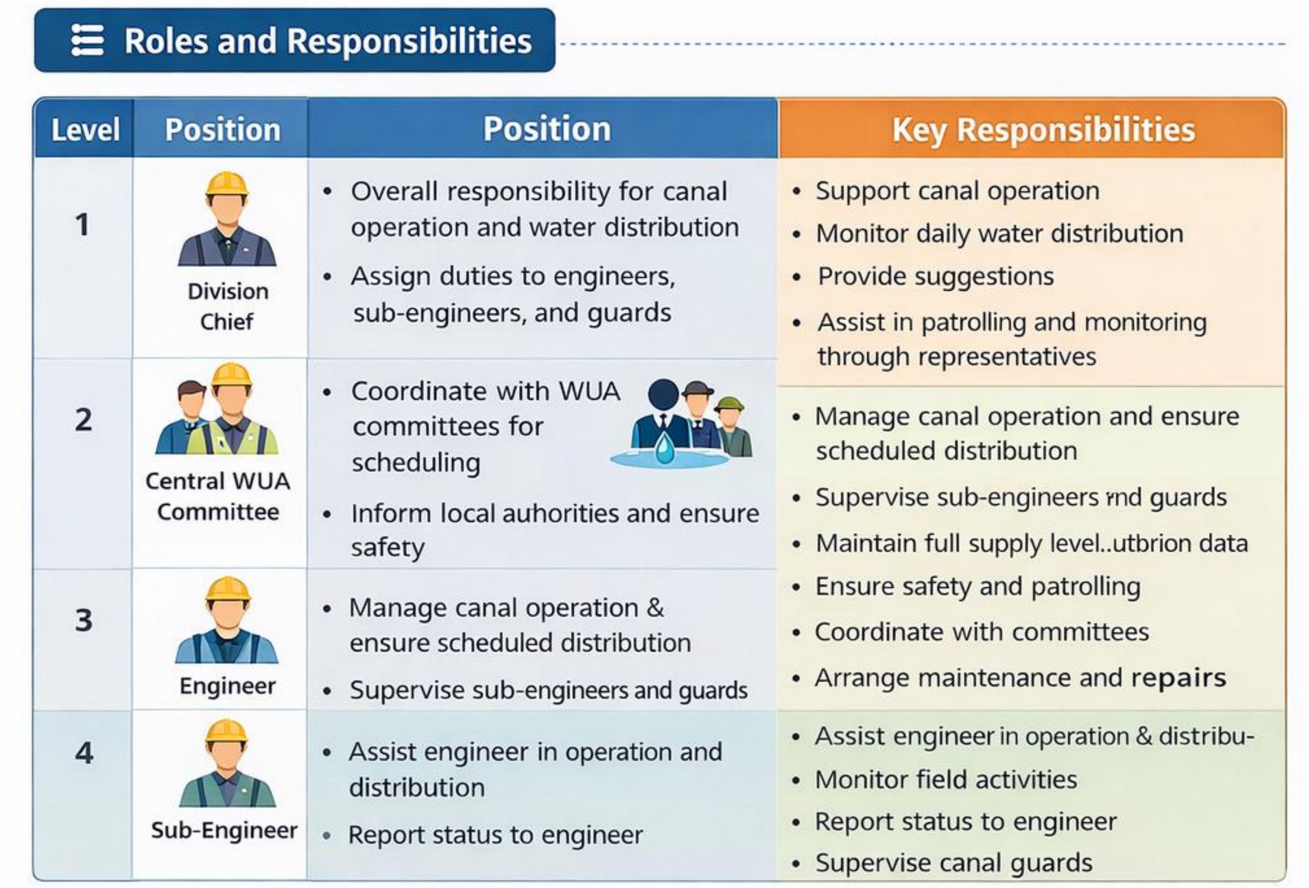
4. Duties of the Sub-Engineer

- To assist in canal operation and water distribution under the guidance of the Engineer.
- To keep the Engineer informed about canal operation and water distribution status.
- To supervise canal guards and provide necessary instructions.

5. Duties of Canal Guards

- To measure and record daily the discharge and duration of water received at the headworks.
- To operate canals and distribute water to branch and main canals as per instructions and schedule.
- To maintain daily records of water distribution.

- To report daily water data to responsible staff at the division office.
- To operate gates and conduct regular patrolling of assigned canal sections.
- To remove weeds and vegetation from canals and irrigation structures to maintain efficiency and aesthetics.
- To carry out minor repairs as instructed.
- To promptly report any problems observed in the canal to Engineers or Sub-engineers.



Roles and Responsibilities

Level	Position	Position	Key Responsibilities
1	Division Chief	• Overall responsibility for canal operation and water distribution • Assign duties to engineers, sub-engineers, and guards	• Support canal operation • Monitor daily water distribution • Provide suggestions • Assist in patrolling and monitoring through representatives
2	Central WUA Committee	• Coordinate with WUA committees for scheduling • Inform local auhorities and ensure safety	• Manage canal operation and ensure scheduled distribution • Supervise sub-engineers rnd guards • Maintain full supply level..utbrion data
3	Engineer	• Manage canal operation & ensure scheduled distribution • Supervise sub-engineers and guards	• Ensure safety and patrolling • Coordinate with committees • Arrange maintenance and repairs
4	Sub-Engineer	• Assist engineer in operation and distribution • Report status to engineer	• Assist engineer in operation & distribu- • Monitor field activities • Report status to engineer • Supervise canal guards

Figure 23 : Key responsibilities based on relative positions

7 Canal Operation Procedures

Canal operation procedures are established to ensure the efficient, equitable, and sustainable distribution of irrigation water throughout the command area of both Eastern and Western canal system. These procedures include planning, regulation, monitoring, communication, and emergency response activities that help maintain smooth canal operations and meet the irrigation demands of beneficiary farmers.

7.1 Pre-Season Preparation

Before the irrigation season begins, both eastern and western canal systems must be inspected and prepared for operation. This includes cleaning canals, removing sediment and vegetation, repairing damaged structures, checking gates and control devices, and assessing water availability. Staff should also review operation schedules, coordinate with Water Users Associations (WUAs), and finalize water distribution plans to ensure readiness for the upcoming season.

7.2 Day-to-Day Canal Operation Techniques

Daily canal operations involve regulating water flow according to approved schedules and crop water requirements. Operators should monitor canal discharge, adjust gates, inspect structures, and address operational issues to maintain efficient water delivery. Regular field observations and record keeping help ensure equitable distribution and minimize water losses throughout the irrigation network.

7.3 Communication and Coordination Protocols

Effective communication should be maintained among Kamala irrigation system officials, canal operators, Water Users Associations, and farmers. Regular meetings, field visits, and information sharing mechanisms should be used to coordinate water delivery schedules and resolve operational issues. This timely communication helps improve the major issue in this irrigation system of reducing conflicts, and ensuring efficient management of irrigation services.

7.4 Emergency and Special Operation Procedures

Emergency procedures should be implemented during floods, droughts, canal breaches, equipment failures, or any other types of unforeseen events. Immediate actions include regulating water flow, protecting canal structures, notifying every stakeholder, and mobilizing repair teams. Special operating arrangements may also be adopted during periods of water scarcity or excessive demand to ensure safe and equitable water distribution.

7.5 Water Distribution Regulations and Enforcement

Water distribution is carried out according to approved operational rules, allocation schedules, and irrigation requirements. Canal operators and management authorities monitor compliance with these regulations to prevent unauthorized water use and wastage. Enforcement measures may include inspections, warnings, and coordination with Water Users Associations to ensure fair water allocation among all beneficiaries.

8 Recommendations and Conclusions

8.1 Recommendations for performance enhancement

8.2 Key Recommendations and Conclusions

Annexes: Annex I: WUA Organizational details

Kamala Eastern Canal -Committee Member			
S.N.	**Post**	**Name**	**Gender**
1	President	Ram Udgar Yadav	Male
2	Vice-President	Kedar Prasad Koirala	Male
3	Secretary	Hemani Kumari Yadav	Female
4	Treasurer	Ram Balak Yadav	Male
5	Assistant Secretary	Manju Devi Yadav	Female
6	Committee Member	Prasila Yadav	Female
7	Committee Member	Ram Udgar Yadav	Male
8	Committee Member	Saukhi Yadav	Female
9	Committee Member	Jitendra Yadav	Male
10	Committee Member	Narad Lal Shrestha	Male
11	Committee Member	Punam Devi Yadav	Female
12	Committee Member	Dhanik Lal Yadav	Male
13	Committee Member	Lalan Kumar Yadav	Male
14	Committee Member	Bhogendra Paswan	Male
15	Committee Member	Ramji Devi Barahi	Female
16	Committee Member	Ram Gulam Kapar	Male
17	Committee Member	Inadala Kumari Yadav	Female
18	Committee Member	Ranjita Devi Yadav	Female
19	Committee Member	Sukila Kumari Yadav	Female
20	Committee Member	Rainul Khatun	Male
21	Committee Member	Mahesh mandal	Male

Kamala Western Canal -Committee Member			
S.N.	**Designation**	**Name**	**Gender**
1	President	Shyam Babu Yadav	M
2	Vice-President	Sanjeev Yadav	M
3	Secretary	Jagadish Yadav	M
4	Treasurer	Barun Kumar Yadav	M
5	Assistant Secretary	RamJyoti Yadav	F
6	Committee Member	Adhiti Devi Gareli	F
7	Committee Member	Pradip Yadav	M
8	Committee Member	Ramchandra Yadav	M
9	Committee Member	Shesh Narayan Yadav	M
10	Committee Member	Hari Kishor Yadav	M
11	Committee Member	Jibchi Shah	F
12	Committee Member	Bharat Yadav	M
13	Committee Member	Ramdev Shah Tali	M
14	Committee Member	Jibchi Devi mandal	F
15	Committee Member	JItendra Yadav	M
16	Committee Member	Shikendra Yadav	M
17	Committee Member	Suma Devi Yadav	F

Annex 2: Irrigation scheduling for EMC

Table 27 : Rice irrigation Schedule in 4 batches for EMC

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
1	Bandipur school	1st	9-Jun-26	6:00	10-Jun-26	7:07	25.1	1.0	32	65	20800	20800	230	139.1
2	Bandipur nursery	1st	9-Jun-26	6:00	12-Jun-26	4:50	70.8	3.0	51	65	33150	33150	130	392.3
3	Bandipur 1 kha	1st	9-Jun-26	6:00	9-Jun-26	19:07	13.1	0.5	24	65	15600	15600	330	72.7
4	Bandipur 1 ka	1st	9-Jun-26	6:00	13-Jun-26	18:20	108.3	4.5	300	65	195000	195000	500	600
5	Shiv mandir left	1st	9-Jun-26	6:00	10-Jun-26	2:50	20.8	0.9	15	65	9750	9750	130	115.4
6	Shiv mandir right	1st	9-Jun-26	6:00	9-Jun-26	10:10	4.2	0.2	3	65	1950	1950	130	23.1
7	Tikapur minor	1st	9-Jun-26	6:00	11-Jun-26	14:56	56.9	2.4	41	65	26650	26650	130	315.4
8	Bhulkiya left	1st	9-Jun-26	6:00	12-Jun-26	14:51	80.9	3.4	103	65	66950	66950	230	447.8
9	Bhulkiya right	1st	9-Jun-26	6:00	12-Jun-26	23:50	89.8	3.7	103	65	66950	74389	230	497.6
10	BOKRAHA BC	1st	9-Jun-26	6:00	13-Jun-26	16:12	106.2	4.4	450	65	292500	325000	850	588.2
11	Karjanha 1	1st	9-Jun-26	6:00	10-Jun-26	17:29	35.5	1.5	23	65	14950	16611	130	196.6
12	Karjanha 2	1st	9-Jun-26	6:00	14-Jun-26	7:54	121.9	5.1	79	65	51350	57056	130	675.2
13	Manharwa	1st	9-Jun-26	6:00	14-Jun-26	3:56	117.9	4.9	194	65	126100	140111	330	653.2
14	Birta	1st	9-Jun-26	6:00	11-Jun-26	8:55	50.9	2.1	33	65	21450	23833	130	282.1
15	Malhaniya pirari	1st	9-Jun-26	6:00	14-Jun-26	4:49	118.8	5.0	77	65	50050	55611	130	658.1
16	Gautari	1st	9-Jun-26	6:00	15-Jun-26	17:15	155.3	6.5	178	65	115700	128556	230	859.9
17	SIKRON BC	1st	9-Jun-26	6:00	16-Jun-26	7:03	169.1	7.0	632	65	410800	456444	750	936.3
18	Arang	1st	9-Jun-26	6:00	18-Jun-26	2:46	212.8	8.9	350	65	227500	252778	330	1178.5
19	MAJHAU BC	1st	9-Jun-26	6:00	26-Jun-26	12:57	415.0	17.3	1179	65	766350	851500	570	2298.2
20	Dumri	1st	9-Jun-26	6:00	14-Jun-26	10:43	124.7	5.2	143	65	92950	103278	230	690.8
21	Kalyanpur minor	1st	9-Jun-26	6:00	13-Jun-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
22	Dakaha Minor	1st	9-Jun-26	6:00	13-Jun-26	20:46	110.8	4.6	127	65	82550	91722	230	613.5
23	KALYANPUR BC	1st	9-Jun-26	6:00	13-Jun-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
24	Ramna	1st	9-Jun-26	6:00	11-Jun-26	5:50	47.8	2.0	31	65	20150	22389	130	265

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
25	Radhanagar	1st	9-Jun-26	6:00	11-Jun-26	7:22	49.4	2.1	32	65	20800	23111	130	273.5
26	Rajokhari	1st	9-Jun-26	6:00	19-Jun-26	16:00	250.0	10.4	162	65	105300	117000	130	1384.6
27	SUKHTHOR BC	1st	9-Jun-26	6:00	13-Jun-26	2:17	92.3	3.8	690	65	448500	498333	1500	511.1
28	SIRAHA BC	1st	9-Jun-26	6:00	14-Jun-26	20:34	134.6	5.6	1677	65	1090050	1211167	2500	745.3
29	Laxmipur	1st	9-Jun-26	6:00	19-Jun-26	14:35	248.6	10.4	285	65	185250	205833	230	1376.8
30	Jamuwa	1st	9-Jun-26	6:00	16-Jun-26	7:45	169.8	7.1	110	65	71500	79444	130	940.2
31	Bhalwahi	1st	9-Jun-26	6:00	15-Jun-26	8:36	146.6	6.1	95	65	61750	68611	130	812
32	Larhaiya	1st	9-Jun-26	6:00	17-Jun-26	3:40	189.7	7.9	312	65	202800	225333	330	1050.5
33	Rampur	1st	9-Jun-26	6:00	18-Jun-26	21:08	231.2	9.6	265	65	172250	191389	230	1280.2
34	Barhampuri	1st	9-Jun-26	6:00	13-Jun-26	22:39	112.7	4.7	73	65	47450	52722	130	623.9
35	Arnama	1st	9-Jun-26	6:00	16-Jun-26	15:28	177.5	7.4	115	65	74750	83056	130	982.9
36	LAGDI BC	1st	9-Jun-26	6:00	17-Jun-26	15:49	201.8	8.4	2515	65	1634750	1816389	2500	1117.8
37	Gauripur	1st	9-Jun-26	6:00	15-Jun-26	14:46	152.8	6.4	99	65	64350	71500	130	846.2
38	Hanumannagar	1st	9-Jun-26	6:00	20-Jun-26	15:34	273.6	11.4	450	65	292500	325000	330	1515.2
39	AURAHI BC	1st	9-Jun-26	6:00	18-Jun-26	5:49	215.8	9.0	1405	65	913250	1014722	1306	1195.3
40	Tulsipur	1st	9-Jun-26	6:00	18-Jun-26	8:03	218.1	9.1	250	65	162500	180556	230	1207.7
1	Bandipur school	2nd	24-Jun-26	6:00	25-Jun-26	7:07	25.1	1.0	32	65	20800	20800	230	139.1
2	Bandipur nursery	2nd	24-Jun-26	6:00	27-Jun-26	4:50	70.8	3.0	51	65	33150	33150	130	392.3
3	Bandipur 1 kha	2nd	24-Jun-26	6:00	24-Jun-26	19:07	13.1	0.5	24	65	15600	15600	330	72.7
4	Bandipur 1 ka	2nd	24-Jun-26	6:00	28-Jun-26	18:20	108.3	4.5	300	65	195000	195000	500	600
5	Shiv mandir left	2nd	24-Jun-26	6:00	25-Jun-26	2:50	20.8	0.9	15	65	9750	9750	130	115.4
6	Shiv mandir right	2nd	24-Jun-26	6:00	24-Jun-26	10:10	4.2	0.2	3	65	1950	1950	130	23.1
7	Tikapur minor	2nd	24-Jun-26	6:00	26-Jun-26	14:56	56.9	2.4	41	65	26650	26650	130	315.4
8	Bhulkiya left	2nd	24-Jun-26	6:00	27-Jun-26	14:51	80.9	3.4	103	65	66950	66950	230	447.8
9	Bhulkiya right	2nd	24-Jun-26	6:00	27-Jun-26	23:50	89.8	3.7	103	65	66950	74389	230	497.6
10	BOKRAHA BC	2nd	24-Jun-26	6:00	28-Jun-26	16:12	106.2	4.4	450	65	292500	325000	850	588.2
11	Karjanha 1	2nd	24-Jun-26	6:00	25-Jun-26	17:29	35.5	1.5	23	65	14950	16611	130	196.6
12	Karjanha 2	2nd	24-Jun-26	6:00	29-Jun-26	7:54	121.9	5.1	79	65	51350	57056	130	675.2

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
13	Manharwa	2nd	24-Jun-26	6:00	29-Jun-26	3:56	117.9	4.9	194	65	126100	140111	330	653.2
14	Birta	2nd	24-Jun-26	6:00	26-Jun-26	8:55	50.9	2.1	33	65	21450	23833	130	282.1
15	Malhaniya pirari	2nd	24-Jun-26	6:00	29-Jun-26	4:49	118.8	5.0	77	65	50050	55611	130	658.1
16	Gautari	2nd	24-Jun-26	6:00	30-Jun-26	17:15	155.3	6.5	178	65	115700	128556	230	859.9
17	SIKRON BC	2nd	24-Jun-26	6:00	1-Jul-26	7:03	169.1	7.0	632	65	410800	456444	750	936.3
18	Arang	2nd	24-Jun-26	6:00	3-Jul-26	2:46	212.8	8.9	350	65	227500	252778	330	1178.5
19	MAJHAU BC	2nd	24-Jun-26	6:00	11-Jul-26	12:57	415.0	17.3	1179	65	766350	851500	570	2298.2
20	Dumri	2nd	24-Jun-26	6:00	29-Jun-26	10:43	124.7	5.2	143	65	92950	103278	230	690.8
21	Kalyanpur minor	2nd	24-Jun-26	6:00	28-Jun-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
22	Dakaha Minor	2nd	24-Jun-26	6:00	28-Jun-26	20:46	110.8	4.6	127	65	82550	91722	230	613.5
23	KALYANPUR BC	2nd	24-Jun-26	6:00	28-Jun-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
24	Ramna	2nd	24-Jun-26	6:00	26-Jun-26	5:50	47.8	2.0	31	65	20150	22389	130	265
25	Radhanagar	2nd	24-Jun-26	6:00	26-Jun-26	7:22	49.4	2.1	32	65	20800	23111	130	273.5
26	Rajokhari	2nd	24-Jun-26	6:00	4-Jul-26	16:00	250.0	10.4	162	65	105300	117000	130	1384.6
27	SUKHTHOR BC	2nd	24-Jun-26	6:00	28-Jun-26	2:17	92.3	3.8	690	65	448500	498333	1500	511.1
28	SIRAHA BC	2nd	24-Jun-26	6:00	29-Jun-26	20:34	134.6	5.6	1677	65	1090050	1211167	2500	745.3
29	Laxmipur	2nd	24-Jun-26	6:00	4-Jul-26	14:35	248.6	10.4	285	65	185250	205833	230	1376.8
30	Jamuwa	2nd	24-Jun-26	6:00	1-Jul-26	7:45	169.8	7.1	110	65	71500	79444	130	940.2
31	Bhalwahi	2nd	24-Jun-26	6:00	30-Jun-26	8:36	146.6	6.1	95	65	61750	68611	130	812
32	Larhaiya	2nd	24-Jun-26	6:00	2-Jul-26	3:40	189.7	7.9	312	65	202800	225333	330	1050.5
33	Rampur	2nd	24-Jun-26	6:00	3-Jul-26	21:08	231.2	9.6	265	65	172250	191389	230	1280.2
34	Barhampuri	2nd	24-Jun-26	6:00	28-Jun-26	22:39	112.7	4.7	73	65	47450	52722	130	623.9
35	Arnama	2nd	24-Jun-26	6:00	1-Jul-26	15:28	177.5	7.4	115	65	74750	83056	130	982.9
36	LAGDI BC	2nd	24-Jun-26	6:00	2-Jul-26	15:49	201.8	8.4	2515	65	1634750	1816389	2500	1117.8
37	Gauripur	2nd	24-Jun-26	6:00	30-Jun-26	14:46	152.8	6.4	99	65	64350	71500	130	846.2
38	Hanumannagar	2nd	24-Jun-26	6:00	5-Jul-26	15:34	273.6	11.4	450	65	292500	325000	330	1515.2
39	AURAHI BC	2nd	24-Jun-26	6:00	3-Jul-26	5:49	215.8	9.0	1405	65	913250	1014722	1306	1195.3
40	Tulsipur	2nd	24-Jun-26	6:00	3-Jul-26	8:03	218.1	9.1	250	65	162500	180556	230	1207.7

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
1	Bandipur school	3rd	2-Jul-26	6:00	3-Jul-26	7:07	25.1	1.0	32	65	20800	20800	230	139.1
2	Bandipur nursery	3rd	2-Jul-26	6:00	5-Jul-26	4:50	70.8	3.0	51	65	33150	33150	130	392.3
3	Bandipur 1 kha	3rd	2-Jul-26	6:00	2-Jul-26	19:07	13.1	0.5	24	65	15600	15600	330	72.7
4	Bandipur 1 ka	3rd	2-Jul-26	6:00	6-Jul-26	18:20	108.3	4.5	300	65	195000	195000	500	600
5	Shiv mandir left	3rd	2-Jul-26	6:00	3-Jul-26	2:50	20.8	0.9	15	65	9750	9750	130	115.4
6	Shiv mandir right	3rd	2-Jul-26	6:00	2-Jul-26	10:10	4.2	0.2	3	65	1950	1950	130	23.1
7	Tikapur minor	3rd	2-Jul-26	6:00	4-Jul-26	14:56	56.9	2.4	41	65	26650	26650	130	315.4
8	Bhulkiya left	3rd	2-Jul-26	6:00	5-Jul-26	14:51	80.9	3.4	103	65	66950	66950	230	447.8
9	Bhulkiya right	3rd	2-Jul-26	6:00	5-Jul-26	23:50	89.8	3.7	103	65	66950	74389	230	497.6
10	BOKRAHA BC	3rd	2-Jul-26	6:00	6-Jul-26	16:12	106.2	4.4	450	65	292500	325000	850	588.2
11	Karjanha 1	3rd	2-Jul-26	6:00	3-Jul-26	17:29	35.5	1.5	23	65	14950	16611	130	196.6
12	Karjanha 2	3rd	2-Jul-26	6:00	7-Jul-26	7:54	121.9	5.1	79	65	51350	57056	130	675.2
13	Manharwa	3rd	2-Jul-26	6:00	7-Jul-26	3:56	117.9	4.9	194	65	126100	140111	330	653.2
14	Birta	3rd	2-Jul-26	6:00	4-Jul-26	8:55	50.9	2.1	33	65	21450	23833	130	282.1
15	Malhaniya pirari	3rd	2-Jul-26	6:00	7-Jul-26	4:49	118.8	5.0	77	65	50050	55611	130	658.1
16	Gautari	3rd	2-Jul-26	6:00	8-Jul-26	17:15	155.3	6.5	178	65	115700	128556	230	859.9
17	SIKRON BC	3rd	2-Jul-26	6:00	9-Jul-26	7:03	169.1	7.0	632	65	410800	456444	750	936.3
18	Arang	3rd	2-Jul-26	6:00	11-Jul-26	2:46	212.8	8.9	350	65	227500	252778	330	1178.5
19	MAJHAU BC	3rd	2-Jul-26	6:00	19-Jul-26	12:57	415.0	17.3	1179	65	766350	851500	570	2298.2
20	Dumri	3rd	2-Jul-26	6:00	7-Jul-26	10:43	124.7	5.2	143	65	92950	103278	230	690.8
21	Kalyanpur minor	3rd	2-Jul-26	6:00	6-Jul-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
22	Dakaha Minor	3rd	2-Jul-26	6:00	6-Jul-26	20:46	110.8	4.6	127	65	82550	91722	230	613.5
23	KALYANPUR BC	3rd	2-Jul-26	6:00	6-Jul-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
24	Ramna	3rd	2-Jul-26	6:00	4-Jul-26	5:50	47.8	2.0	31	65	20150	22389	130	265
25	Radhanagar	3rd	2-Jul-26	6:00	4-Jul-26	7:22	49.4	2.1	32	65	20800	23111	130	273.5
26	Rajokhari	3rd	2-Jul-26	6:00	12-Jul-26	16:00	250.0	10.4	162	65	105300	117000	130	1384.6
27	SUKHTHOR BC	3rd	2-Jul-26	6:00	6-Jul-26	2:17	92.3	3.8	690	65	448500	498333	1500	511.1
28	SIRAHA BC	3rd	2-Jul-26	6:00	7-Jul-26	20:34	134.6	5.6	1677	65	1090050	1211167	2500	745.3

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
29	Laxmipur	3rd	2-Jul-26	6:00	12-Jul-26	14:35	248.6	10.4	285	65	185250	205833	230	1376.8
30	Jamuwa	3rd	2-Jul-26	6:00	9-Jul-26	7:45	169.8	7.1	110	65	71500	79444	130	940.2
31	Bhalwahi	3rd	2-Jul-26	6:00	8-Jul-26	8:36	146.6	6.1	95	65	61750	68611	130	812
32	Larhaiya	3rd	2-Jul-26	6:00	10-Jul-26	3:40	189.7	7.9	312	65	202800	225333	330	1050.5
33	Rampur	3rd	2-Jul-26	6:00	11-Jul-26	21:08	231.2	9.6	265	65	172250	191389	230	1280.2
34	Barhampuri	3rd	2-Jul-26	6:00	6-Jul-26	22:39	112.7	4.7	73	65	47450	52722	130	623.9
35	Arnama	3rd	2-Jul-26	6:00	9-Jul-26	15:28	177.5	7.4	115	65	74750	83056	130	982.9
36	LAGDI BC	3rd	2-Jul-26	6:00	10-Jul-26	15:49	201.8	8.4	2515	65	1634750	1816389	2500	1117.8
37	Gauripur	3rd	2-Jul-26	6:00	8-Jul-26	14:46	152.8	6.4	99	65	64350	71500	130	846.2
38	Hanumannagar	3rd	2-Jul-26	6:00	13-Jul-26	15:34	273.6	11.4	450	65	292500	325000	330	1515.2
39	AURAHI BC	3rd	2-Jul-26	6:00	11-Jul-26	5:49	215.8	9.0	1405	65	913250	1014722	1306	1195.3
40	Tulsipur	3rd	2-Jul-26	6:00	11-Jul-26	8:03	218.1	9.1	250	65	162500	180556	230	1207.7
1	Bandipur school	4th	14-Oct-26	6:00	15-Oct-26	7:07	25.1	1.0	32	65	20800	20800	230	139.1
2	Bandipur nursery	4th	14-Oct-26	6:00	17-Oct-26	4:50	70.8	3.0	51	65	33150	33150	130	392.3
3	Bandipur 1 kha	4th	14-Oct-26	6:00	14-Oct-26	19:07	13.1	0.5	24	65	15600	15600	330	72.7
4	Bandipur 1 ka	4th	14-Oct-26	6:00	18-Oct-26	18:20	108.3	4.5	300	65	195000	195000	500	600
5	Shiv mandir left	4th	14-Oct-26	6:00	15-Oct-26	2:50	20.8	0.9	15	65	9750	9750	130	115.4
6	Shiv mandir right	4th	14-Oct-26	6:00	14-Oct-26	10:10	4.2	0.2	3	65	1950	1950	130	23.1
7	Tikapur minor	4th	14-Oct-26	6:00	16-Oct-26	14:56	56.9	2.4	41	65	26650	26650	130	315.4
8	Bhulkiya left	4th	14-Oct-26	6:00	17-Oct-26	14:51	80.9	3.4	103	65	66950	66950	230	447.8
9	Bhulkiya right	4th	14-Oct-26	6:00	17-Oct-26	23:50	89.8	3.7	103	65	66950	74389	230	497.6
10	BOKRAHA BC	4th	14-Oct-26	6:00	18-Oct-26	16:12	106.2	4.4	450	65	292500	325000	850	588.2
11	Karjanha 1	4th	14-Oct-26	6:00	15-Oct-26	17:29	35.5	1.5	23	65	14950	16611	130	196.6
12	Karjanha 2	4th	14-Oct-26	6:00	19-Oct-26	7:54	121.9	5.1	79	65	51350	57056	130	675.2
13	Manharwa	4th	14-Oct-26	6:00	19-Oct-26	3:56	117.9	4.9	194	65	126100	140111	330	653.2
14	Birta	4th	14-Oct-26	6:00	16-Oct-26	8:55	50.9	2.1	33	65	21450	23833	130	282.1
15	Malhaniya pirari	4th	14-Oct-26	6:00	19-Oct-26	4:49	118.8	5.0	77	65	50050	55611	130	658.1
16	Gautari	4th	14-Oct-26	6:00	20-Oct-26	17:15	155.3	6.5	178	65	115700	128556	230	859.9

S N.	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	Cmd Area (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
17	SIKRON BC	4th	14-Oct-26	6:00	21-Oct-26	7:03	169.1	7.0	632	65	410800	456444	750	936.3
18	Arang	4th	14-Oct-26	6:00	23-Oct-26	2:46	212.8	8.9	350	65	227500	252778	330	1178.5
19	MAJHAU BC	4th	14-Oct-26	6:00	31-Oct-26	12:57	415.0	17.3	1179	65	766350	851500	570	2298.2
20	Dumri	4th	14-Oct-26	6:00	19-Oct-26	10:43	124.7	5.2	143	65	92950	103278	230	690.8
21	Kalyanpur minor	4th	14-Oct-26	6:00	18-Oct-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
22	Dakaha Minor	4th	14-Oct-26	6:00	18-Oct-26	20:46	110.8	4.6	127	65	82550	91722	230	613.5
23	KALYANPUR BC	4th	14-Oct-26	6:00	18-Oct-26	20:45	110.8	4.6	911	65	592150	657944	1650	613.5
24	Ramna	4th	14-Oct-26	6:00	16-Oct-26	5:50	47.8	2.0	31	65	20150	22389	130	265
25	Radhanagar	4th	14-Oct-26	6:00	16-Oct-26	7:22	49.4	2.1	32	65	20800	23111	130	273.5
26	Rajokhari	4th	14-Oct-26	6:00	24-Oct-26	16:00	250.0	10.4	162	65	105300	117000	130	1384.6
27	SUKHTHOR BC	4th	14-Oct-26	6:00	18-Oct-26	2:17	92.3	3.8	690	65	448500	498333	1500	511.1
28	SIRAHA BC	4th	14-Oct-26	6:00	19-Oct-26	20:34	134.6	5.6	1677	65	1090050	1211167	2500	745.3
29	Laxmipur	4th	14-Oct-26	6:00	24-Oct-26	14:35	248.6	10.4	285	65	185250	205833	230	1376.8
30	Jamuwa	4th	14-Oct-26	6:00	21-Oct-26	7:45	169.8	7.1	110	65	71500	79444	130	940.2
31	Bhalwahi	4th	14-Oct-26	6:00	20-Oct-26	8:36	146.6	6.1	95	65	61750	68611	130	812
32	Larhaiya	4th	14-Oct-26	6:00	22-Oct-26	3:40	189.7	7.9	312	65	202800	225333	330	1050.5
33	Rampur	4th	14-Oct-26	6:00	23-Oct-26	21:08	231.2	9.6	265	65	172250	191389	230	1280.2
34	Barhampuri	4th	14-Oct-26	6:00	18-Oct-26	22:39	112.7	4.7	73	65	47450	52722	130	623.9
35	Arnama	4th	14-Oct-26	6:00	21-Oct-26	15:28	177.5	7.4	115	65	74750	83056	130	982.9
36	LAGDI BC	4th	14-Oct-26	6:00	22-Oct-26	15:49	201.8	8.4	2515	65	1634750	1816389	2500	1117.8
37	Gauripur	4th	14-Oct-26	6:00	20-Oct-26	14:46	152.8	6.4	99	65	64350	71500	130	846.2
38	Hanumannagar	4th	14-Oct-26	6:00	25-Oct-26	15:34	273.6	11.4	450	65	292500	325000	330	1515.2
39	AURAHI BC	4th	14-Oct-26	6:00	23-Oct-26	5:49	215.8	9.0	1405	65	913250	1014722	1306	1195.3
40	Tulsipur	4th	14-Oct-26	6:00	23-Oct-26	8:03	218.1	9.1	250	65	162500	180556	230	1207.7

Annex 3: Irrigation scheduling for WMC

Table 28 : Rice irrigation Schedule in 4 batches for WMC

Sn	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	CA (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
1	Portaha Labatoli east minor 1	1st	9-Jun-26	6:00	11-Jun-26	18:11	60.2	2.5	45	65	29250	32500	150	333.3
2	Portaha labatoli east minor 2	1st	9-Jun-26	6:00	12-Jun-26	2:12	68.2	2.8	51	65	33150	36833	150	377.8
3	Chitaha east minor 1	1st	9-Jun-26	6:00	9-Jun-26	23:23	17.4	0.7	13	65	8450	9389	150	96.3
4	Chitaha east minor 2	1st	9-Jun-26	6:00	12-Jun-26	0:52	66.9	2.8	50	65	32500	36111	150	370.4
5	Targachi east minor	1st	9-Jun-26	6:00	11-Jun-26	20:51	62.9	2.6	47	65	30550	33944	150	348.1
6	Raghunathpur Branch	1st	9-Jun-26	6:00	15-Jun-26	13:09	151.2	6.3	1507	65	979550	1088389	2000	837.2
7	Barmajhiya chowk east minor	1st	9-Jun-26	6:00	10-Jun-26	2:03	20.1	0.8	10	65	6500	7222	100	111.1
8	Jitan west minor	1st	9-Jun-26	6:00	9-Jun-26	20:02	14.0	0.6	7	65	4550	5056	100	77.8
9	Barmajhiya east minor 1	1st	9-Jun-26	6:00	9-Jun-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
10	Barmajhiya east minor 2	1st	9-Jun-26	6:00	13-Jun-26	18:20	108.3	4.5	108	65	70200	78000	200	600
11	Parwaha Branch	1st	9-Jun-26	6:00	15-Jun-26	16:46	154.8	6.4	2427	65	1577550	1752833	3146	857.2
12	Dumariya east minor 1	1st	9-Jun-26	6:00	10-Jun-26	14:05	32.1	1.3	16	65	10400	11556	100	177.8
13	Dumariya east minor 2	1st	9-Jun-26	6:00	13-Jun-26	11:54	101.9	4.2	127	65	82550	91722	250	564.4
14	Dumariya east minor ka	1st	9-Jun-26	6:00	12-Jun-26	10:14	76.2	3.2	38	65	24700	27444	100	422.2
15	Sabela west minor 1	1st	9-Jun-26	6:00	14-Jun-26	8:05	122.1	5.1	213	65	138450	153833	350	676.2
16	Sabela west minor 2	1st	9-Jun-26	6:00	14-Jun-26	1:21	115.4	4.8	115	65	74750	83056	200	638.9
17	Kothiya minor east	1st	9-Jun-26	6:00	15-Jun-26	6:26	144.4	6.0	72	65	46800	52000	100	800
18	Kothiya west minor ka	1st	9-Jun-26	6:00	9-Jun-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
19	Barkurwa east minor	1st	9-Jun-26	6:00	16-Jun-26	10:31	172.5	7.2	86	65	55900	62111	100	955.6
20	Bagwan east minor	1st	9-Jun-26	6:00	15-Jun-26	9:53	147.9	6.2	258	65	167700	186333	350	819
21	Bagwan east minor ka	1st	9-Jun-26	6:00	14-Jun-26	12:23	126.4	5.3	63	65	40950	45500	100	700
22	Thilla west minor 1	1st	9-Jun-26	6:00	9-Jun-26	10:00	4.0	0.2	2	65	1300	1444	100	22.2
23	Kulwamod east minor	1st	9-Jun-26	6:00	13-Jun-26	2:17	92.3	3.8	115	65	74750	83056	250	511.1
24	Thilla west minor 2	1st	9-Jun-26	6:00	10-Jun-26	0:03	18.1	0.8	9	65	5850	6500	100	100
25	Thilla west minor 3	1st	9-Jun-26	6:00	11-Jun-26	10:09	52.2	2.2	26	65	16900	18778	100	288.9

Sn	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	CA (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
26	Kharihani west minor	1st	9-Jun-26	6:00	12-Jun-26	15:51	81.9	3.4	102	65	66300	73667	250	453.3
27	Khajuri Branch	1st	9-Jun-26	6:00	15-Jun-26	18:00	156.0	6.5	3570	65	2320500	2578333	4591	864
28	Mahinathpur Branch	1st	9-Jun-26	6:00	14-Jun-26	5:03	119.1	5.0	1968	65	1279200	1421333	3316	659.4
29	Portaha Labatoli east minor 1	2nd	24-Jun-26	6:00	26-Jun-26	18:11	60.2	2.5	45	65	29250	32500	150	333.3
30	Portaha labatoli east minor 2	2nd	24-Jun-26	6:00	27-Jun-26	2:12	68.2	2.8	51	65	33150	36833	150	377.8
31	Chitaha east minor 1	2nd	24-Jun-26	6:00	24-Jun-26	23:23	17.4	0.7	13	65	8450	9389	150	96.3
32	Chitaha east minor 2	2nd	24-Jun-26	6:00	27-Jun-26	0:52	66.9	2.8	50	65	32500	36111	150	370.4
33	Targachi east minor	2nd	24-Jun-26	6:00	26-Jun-26	20:51	62.9	2.6	47	65	30550	33944	150	348.1
34	Raghunathpur Branch	2nd	24-Jun-26	6:00	30-Jun-26	13:09	151.2	6.3	1507	65	979550	1088389	2000	837.2
35	Barmajhiya chowk east minor	2nd	24-Jun-26	6:00	25-Jun-26	2:03	20.1	0.8	10	65	6500	7222	100	111.1
36	Jitan west minor	2nd	24-Jun-26	6:00	24-Jun-26	20:02	14.0	0.6	7	65	4550	5056	100	77.8
37	Barmajhiya east minor 1	2nd	24-Jun-26	6:00	24-Jun-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
38	Barmajhiya east minor 2	2nd	24-Jun-26	6:00	28-Jun-26	18:20	108.3	4.5	108	65	70200	78000	200	600
39	Parwaha Branch	2nd	24-Jun-26	6:00	30-Jun-26	16:46	154.8	6.4	2427	65	1577550	1752833	3146	857.2
40	Dumariya east minor 1	2nd	24-Jun-26	6:00	25-Jun-26	14:05	32.1	1.3	16	65	10400	11556	100	177.8
41	Dumariya east minor 2	2nd	24-Jun-26	6:00	28-Jun-26	11:54	101.9	4.2	127	65	82550	91722	250	564.4
42	Dumariya east minor ka	2nd	24-Jun-26	6:00	27-Jun-26	10:14	76.2	3.2	38	65	24700	27444	100	422.2
43	Sabela west minor 1	2nd	24-Jun-26	6:00	29-Jun-26	8:05	122.1	5.1	213	65	138450	153833	350	676.2
44	Sabela west minor 2	2nd	24-Jun-26	6:00	29-Jun-26	1:21	115.4	4.8	115	65	74750	83056	200	638.9
45	Kothiya minor east	2nd	24-Jun-26	6:00	30-Jun-26	6:26	144.4	6.0	72	65	46800	52000	100	800
46	Kothiya west minor ka	2nd	24-Jun-26	6:00	24-Jun-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
47	Barkurwa east minor	2nd	24-Jun-26	6:00	1-Jul-26	10:31	172.5	7.2	86	65	55900	62111	100	955.6
48	Bagwan east minor	2nd	24-Jun-26	6:00	30-Jun-26	9:53	147.9	6.2	258	65	167700	186333	350	819
49	Bagwan east minor ka	2nd	24-Jun-26	6:00	29-Jun-26	12:23	126.4	5.3	63	65	40950	45500	100	700
50	Thilla west minor 1	2nd	24-Jun-26	6:00	24-Jun-26	10:00	4.0	0.2	2	65	1300	1444	100	22.2
51	Kulwamod east minor	2nd	24-Jun-26	6:00	28-Jun-26	2:17	92.3	3.8	115	65	74750	83056	250	511.1
52	Thilla west minor 2	2nd	24-Jun-26	6:00	25-Jun-26	0:03	18.1	0.8	9	65	5850	6500	100	100
53	Thilla west minor 3	2nd	24-Jun-26	6:00	26-Jun-26	10:09	52.2	2.2	26	65	16900	18778	100	288.9

Sn	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	CA (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
54	Kharihani west minor	2nd	24-Jun-26	6:00	27-Jun-26	15:51	81.9	3.4	102	65	66300	73667	250	453.3
55	Khajuri Branch	2nd	24-Jun-26	6:00	30-Jun-26	18:00	156.0	6.5	3570	65	2320500	2578333	4591	864
56	Mahinathpur Branch	2nd	24-Jun-26	6:00	29-Jun-26	5:03	119.1	5.0	1968	65	1279200	1421333	3316	659.4
57	Portaha Labatoli east minor 1	3rd	2-Jul-26	6:00	4-Jul-26	18:11	60.2	2.5	45	65	29250	32500	150	333.3
58	Portaha labatoli east minor 2	3rd	2-Jul-26	6:00	5-Jul-26	2:12	68.2	2.8	51	65	33150	36833	150	377.8
59	Chitaha east minor 1	3rd	2-Jul-26	6:00	2-Jul-26	23:23	17.4	0.7	13	65	8450	9389	150	96.3
60	Chitaha east minor 2	3rd	2-Jul-26	6:00	5-Jul-26	0:52	66.9	2.8	50	65	32500	36111	150	370.4
61	Targachi east minor	3rd	2-Jul-26	6:00	4-Jul-26	20:51	62.9	2.6	47	65	30550	33944	150	348.1
62	Raghunathpur Branch	3rd	2-Jul-26	6:00	8-Jul-26	13:09	151.2	6.3	1507	65	979550	1088389	2000	837.2
63	Barmajhiya chowk east minor	3rd	2-Jul-26	6:00	3-Jul-26	2:03	20.1	0.8	10	65	6500	7222	100	111.1
64	Jitan west minor	3rd	2-Jul-26	6:00	2-Jul-26	20:02	14.0	0.6	7	65	4550	5056	100	77.8
65	Barmajhiya east minor 1	3rd	2-Jul-26	6:00	2-Jul-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
66	Barmajhiya east minor 2	3rd	2-Jul-26	6:00	6-Jul-26	18:20	108.3	4.5	108	65	70200	78000	200	600
67	Parwaha Branch	3rd	2-Jul-26	6:00	8-Jul-26	16:46	154.8	6.4	2427	65	1577550	1752833	3146	857.2
68	Dumariya east minor 1	3rd	2-Jul-26	6:00	3-Jul-26	14:05	32.1	1.3	16	65	10400	11556	100	177.8
69	Dumariya east minor 2	3rd	2-Jul-26	6:00	6-Jul-26	11:54	101.9	4.2	127	65	82550	91722	250	564.4
70	Dumariya east minor ka	3rd	2-Jul-26	6:00	5-Jul-26	10:14	76.2	3.2	38	65	24700	27444	100	422.2
71	Sabela west minor 1	3rd	2-Jul-26	6:00	7-Jul-26	8:05	122.1	5.1	213	65	138450	153833	350	676.2
72	Sabela west minor 2	3rd	2-Jul-26	6:00	7-Jul-26	1:21	115.4	4.8	115	65	74750	83056	200	638.9
73	Kothiya minor east	3rd	2-Jul-26	6:00	8-Jul-26	6:26	144.4	6.0	72	65	46800	52000	100	800
74	Kothiya west minor ka	3rd	2-Jul-26	6:00	2-Jul-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
75	Barkurwa east minor	3rd	2-Jul-26	6:00	9-Jul-26	10:31	172.5	7.2	86	65	55900	62111	100	955.6
76	Bagwan east minor	3rd	2-Jul-26	6:00	8-Jul-26	9:53	147.9	6.2	258	65	167700	186333	350	819
77	Bagwan east minor ka	3rd	2-Jul-26	6:00	7-Jul-26	12:23	126.4	5.3	63	65	40950	45500	100	700
78	Thilla west minor 1	3rd	2-Jul-26	6:00	2-Jul-26	10:00	4.0	0.2	2	65	1300	1444	100	22.2
79	Kulwamod east minor	3rd	2-Jul-26	6:00	6-Jul-26	2:17	92.3	3.8	115	65	74750	83056	250	511.1
80	Thilla west minor 2	3rd	2-Jul-26	6:00	3-Jul-26	0:03	18.1	0.8	9	65	5850	6500	100	100
81	Thilla west minor 3	3rd	2-Jul-26	6:00	4-Jul-26	10:09	52.2	2.2	26	65	16900	18778	100	288.9

Sn	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	CA (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
82	Kharihani west minor	3rd	2-Jul-26	6:00	5-Jul-26	15:51	81.9	3.4	102	65	66300	73667	250	453.3
83	Khajuri Branch	3rd	2-Jul-26	6:00	8-Jul-26	18:00	156.0	6.5	3570	65	2320500	2578333	4591	864
84	Mahinathpur Branch	3rd	2-Jul-26	6:00	7-Jul-26	5:03	119.1	5.0	1968	65	1279200	1421333	3316	659.4
85	Portaha Labatoli east minor 1	4th	14-Oct-26	6:00	16-Oct-26	18:11	60.2	2.5	45	65	29250	32500	150	333.3
86	Portaha labatoli east minor 2	4th	14-Oct-26	6:00	17-Oct-26	2:12	68.2	2.8	51	65	33150	36833	150	377.8
87	Chitaha east minor 1	4th	14-Oct-26	6:00	14-Oct-26	23:23	17.4	0.7	13	65	8450	9389	150	96.3
88	Chitaha east minor 2	4th	14-Oct-26	6:00	17-Oct-26	0:52	66.9	2.8	50	65	32500	36111	150	370.4
89	Targachi east minor	4th	14-Oct-26	6:00	16-Oct-26	20:51	62.9	2.6	47	65	30550	33944	150	348.1
90	Raghunathpur Branch	4th	14-Oct-26	6:00	20-Oct-26	13:09	151.2	6.3	1507	65	979550	1088389	2000	837.2
91	Barmajhiya chowk east minor	4th	14-Oct-26	6:00	15-Oct-26	2:03	20.1	0.8	10	65	6500	7222	100	111.1
92	Jitan west minor	4th	14-Oct-26	6:00	14-Oct-26	20:02	14.0	0.6	7	65	4550	5056	100	77.8
93	Barmajhiya east minor 1	4th	14-Oct-26	6:00	14-Oct-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
94	Barmajhiya east minor 2	4th	14-Oct-26	6:00	18-Oct-26	18:20	108.3	4.5	108	65	70200	78000	200	600
95	Parwaha Branch	4th	14-Oct-26	6:00	20-Oct-26	16:46	154.8	6.4	2427	65	1577550	1752833	3146	857.2
96	Dumariya east minor 1	4th	14-Oct-26	6:00	15-Oct-26	14:05	32.1	1.3	16	65	10400	11556	100	177.8
97	Dumariya east minor 2	4th	14-Oct-26	6:00	18-Oct-26	11:54	101.9	4.2	127	65	82550	91722	250	564.4
98	Dumariya east minor ka	4th	14-Oct-26	6:00	17-Oct-26	10:14	76.2	3.2	38	65	24700	27444	100	422.2
99	Sabela west minor 1	4th	14-Oct-26	6:00	19-Oct-26	8:05	122.1	5.1	213	65	138450	153833	350	676.2
100	Sabela west minor 2	4th	14-Oct-26	6:00	19-Oct-26	1:21	115.4	4.8	115	65	74750	83056	200	638.9
101	Kothiya minor east	4th	14-Oct-26	6:00	20-Oct-26	6:26	144.4	6.0	72	65	46800	52000	100	800
102	Kothiya west minor ka	4th	14-Oct-26	6:00	14-Oct-26	22:02	16.1	0.7	8	65	5200	5778	100	88.9
103	Barkurwa east minor	4th	14-Oct-26	6:00	21-Oct-26	10:31	172.5	7.2	86	65	55900	62111	100	955.6
104	Bagwan east minor	4th	14-Oct-26	6:00	20-Oct-26	9:53	147.9	6.2	258	65	167700	186333	350	819
105	Bagwan east minor ka	4th	14-Oct-26	6:00	19-Oct-26	12:23	126.4	5.3	63	65	40950	45500	100	700
106	Thilla west minor 1	4th	14-Oct-26	6:00	14-Oct-26	10:00	4.0	0.2	2	65	1300	1444	100	22.2
107	Kulwamod east minor	4th	14-Oct-26	6:00	18-Oct-26	2:17	92.3	3.8	115	65	74750	83056	250	511.1
108	Thilla west minor 2	4th	14-Oct-26	6:00	15-Oct-26	0:03	18.1	0.8	9	65	5850	6500	100	100
109	Thilla west minor 3	4th	14-Oct-26	6:00	16-Oct-26	10:09	52.2	2.2	26	65	16900	18778	100	288.9

Sn	Canal Name	Batch	Start Date	Start Time	End Date	End Time	Duration (hrs)	Duration (days)	CA (Ha)	Depth (mm)	Net Volume (m3)	Gross Volume (m3)	Flow (lps)	Duty (Ha/cumec)
110	Kharihani west minor	4th	14-Oct-26	6:00	17-Oct-26	15:51	81.9	3.4	102	65	66300	73667	250	453.3
111	Khajuri Branch	4th	14-Oct-26	6:00	20-Oct-26	18:00	156.0	6.5	3570	65	2320500	2578333	4591	864
112	Mahinathpur Branch	4th	14-Oct-26	6:00	19-Oct-26	5:03	119.1	5.0	1968	65	1279200	1421333	3316	659.4